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**STUDY OF TAU LEPTON POLARIZATION
WITH THE IDEA DETECTOR AT FCC-ee**

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In God we trust.
All others must bring data.

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Introduction

The Standard Model of particle physics has so far successfully handled any experimental testing and probing, however, it fails to produce predictions for currently open questions, such as the nature of dark energy and dark matter or the neutrino mass scale. The discovery of the Higgs Boson in 2012 and the absence of evident particle phenomena beyond the SM up to energies of $\sqrt{s} = 13$ TeV have sparked interest in future colliders in order to extend the involved energy scales and the precision in the measurements of the model's parameters. High precision measurements could indicate the direction for future extensions of the Standard Model, mainly by producing low-energy corrections to new phenomena that could manifest only at energies much higher than the TeV scale.

The following work studies the experimental precision of the measurement of the parameter $\sin^2 \theta_W$ through the polarization $\mathcal{P}_\tau$ of the τ leptons produced in e^+e^- collisions on the Z resonance. The parameter is studied in a possible configuration of the IDEA detector [1] at the Future Circular Collider (FCC)[2]. The focus of the study is to evaluate the impact of experimental uncertainties and reconstruction efficiencies in this measurement; this is of great importance given the unprecedented statistics which future e^+e^- colliders such as FCC will grant: $\mathcal{O}(10^{12})$ Z boson decays.

The sample used in this study consists of 100 million events of the process

$$e^+e^- \rightarrow Z \rightarrow \tau^+\tau^-$$

which are generated by the Monte Carlo program PYTHIA8 and subsequently propagated through a simulated environment of the IDEA detector using the DELPHES framework.

Chapter 1

The Future Circular Collider

In 2013 the Update of the European Strategy for Particle Physics [3] clearly paved the path to a future of widespread collaborations with great emphasis on a "*vigorous accelerator R&D programme*". As a consequence the proposal for the Future Circular Collider emerged and is now the leading candidate for the future of the particle physics program at CERN.

1.1 The FCC project

The Future Circular Collider is a conceptual design collider, proposed by the European Organization for Nuclear Research (CERN) and is proposed to succeed to the Large Hadron Collider (LHC) and its High-Luminosity program. The project is regularly updated through feasibility studies [4] and includes a ~ 100 km collider ring which will host up to 4 experiments placed around its circumference. The collider aims to scan through a wide spread of center of mass energies ($\sqrt{s}$) in the electro-weak regime and

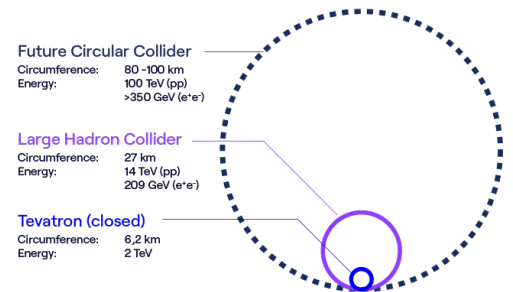


Figure 1.1: Scale of the FCC project [4].

1.1 The FCC project

reach its final 100 TeV goal.

The collider would deliver unprecedented event statistics, the ideal environment of high precision measurements. The operational stages of FCC are divided into 3 main steps, each characterized by different types of particle collisions in the ring:

- FCC-ee [2] (2045-2060) will consist of e^+e^- collisions at very high luminosity, providing access to measurements with an accuracy order(s) of magnitude better than today.
- FCC-hh [5] (2070-x) will provide hadronic collisions which will push the energy frontier to the 100 TeV mark, serving the role of a Higgs factory.
- FCC-eh [6] (undefined) will collide a hadronic beam from FCC-hh with a 60 GeV electron beam recovered from a linear accelerator. The lepton-hadron collision will provide great accuracy in the study of partonic distributions, a crucial step also for the analysis of FCC data from previous runs.

The following study is representative of the FCC-ee phase, in particular its operational stage revolved around the Z pole; the complete plan is reported in table 1.1. The latest update [7] also considers other options for the collider program, in particular its plan B is a *descope*d FCC-ee. In this secondary scenario there would be partial cuts to the program: the $t\bar{t}$ stage would be removed, the number of experimental caverns would be reduced to 2 together with a limitation on the radiofrequency (RF) of the system power. These cuts would yield an overall $\sim 15\%$ budget cut compared to a full FCC-ee, but also reduce the breadth of the physics program.

Phase	(years)	CM energy ($\sqrt{s}$ [GeV])	Integrated Luminosity ($\mathcal{L}_{\text{int}}$ [ab^{-1}])	Event statistics
FCC-ee-Z	4	88–94	205	6×10^{12} Z
FCC-ee-W	2	157–163	19.2	2.4×10^8 WW
FCC-ee-H	3	240	1.08	2.2×10^6 ZH
FCC-ee-tt(1)	1	340–350	0.42	$t\bar{t}$ threshold
FCC-ee-tt(2)	4	365	2.70	2×10^6 $t\bar{t}$ events

Table 1.1: Plan A: phases of operation during FCC-ee [8].

1.2 IDEA detector

One of the detectors proposed for the realization of the FCC project is the IDEA detector. The acronym stands for *The Innovative Detector for Electron-Positron Accelerator*. The following description reports the latest information dated March 2025 [1], however, it should be noted that this thesis used data generated from the 2023 winter simulation production. While there have been changes in the engineering of the project the data should still reflect the overall performance of the IDEA detector.

Some key features of the detector are good momentum, angle and impact parameter resolution. Many experimental aspects drive the detector design, in order to reach the required performances. For example a precise vertex detector, together with good impact parameter precision, allows for an improved efficiency in heavy flavour tagging in jet events. Other sectors, such as Higgs physics, benefit instead from precision measurements of high momentum tracks. The necessary requirement for IDEA then is a precision of $\frac{\sigma_{p_t}}{p_t} \sim 10^{-5}$. Driven by these constraints, and many others, the IDEA design is continuously updated through a vigorous R&D effort by the research group.

The layout of the IDEA detector is represented in figures 1.2 , 1.3 and can be

1.2 IDEA detector

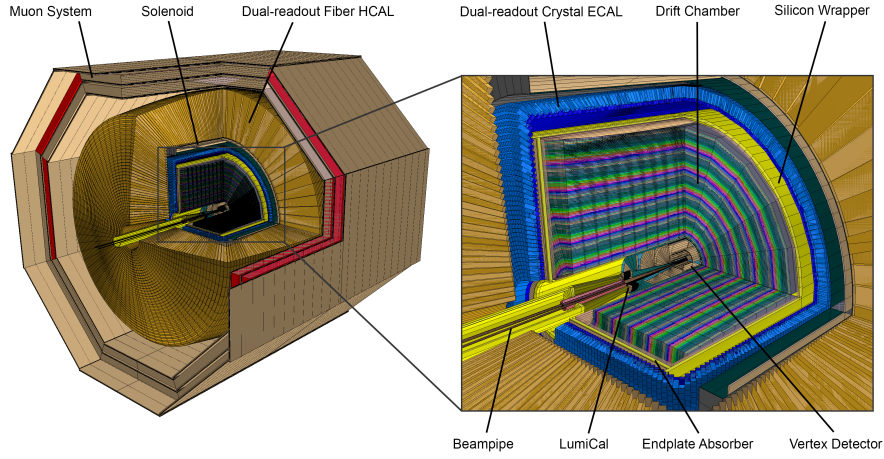


Figure 1.2: 3D cutout view of the IDEA detector [1]

summarized in the following subsections.

Tracking system

The tracking system is composed of a silicon vertex detector, a lightweight drift chamber and an external silicon wrapper. The angular acceptance is around $|\cos \theta| < 0.99$, which provides coverage of (almost) the entire stereo-angle.

The first stage is the high resolution ultra light silicon pixel vertex detector. In order to obtain precise primary and secondary vertices reconstructions it requires μm single digit precision. Different technological and geometrical proposals have been put forward but they mainly make use of monolithic active pixel sensors (MAPS).

Around the vertex stage we find the drift chamber: a single-volume, cylindrical wire chamber designed to provide precise momentum and tracking by implementing a "cluster counting" algorithm. This technique exploits the superior resolution of separation and identification via the number of ionization events per unit length dN/dx compared to the more traditional dE/dx . An interesting peculiarity of the drift chamber is its high transparency which is crucial to minimize multiple

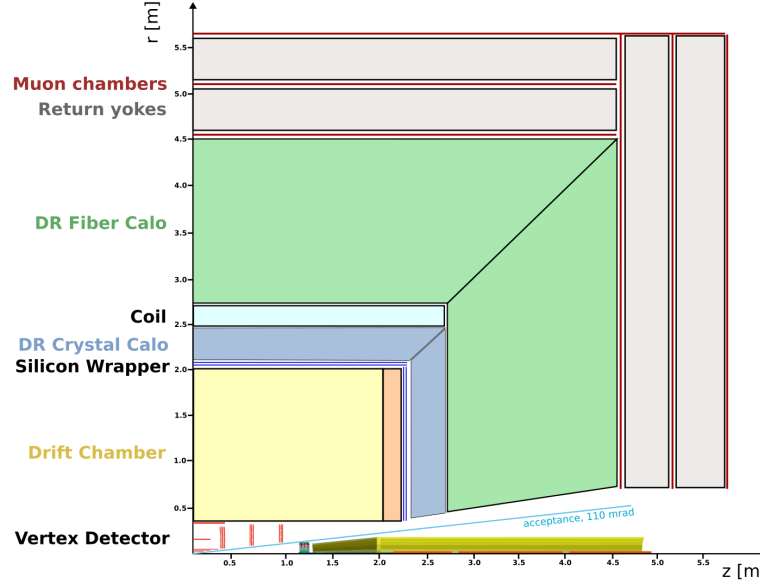


Figure 1.3: Schematic overview and geometrical dimensions of the IDEA detector layout [1]

scatterings and conversions/hadronic interactions.

Located around the drift chamber, the silicon wrapper is the last stage of the tracking system. It is able to provide full hermeticity, enabling the full stereo angle coverage. Different technologies have been proposed with different features and engineering solutions. Given the technological choice, IDEA also aims to achieve $\sim 20 - 100$ ns time of flight measurements, either within this stage of the detector or with the crystal calorimeter.

Calorimetry system

The calorimetry system for IDEA is designed to achieve excellent energy resolution for both electromagnetic and hadronic showers. This ambitious proposal is based on a dual-readout technique obtained with event-by-event corrections in the EM fraction of hadronic showers thanks to the collected scintillation and Cherenkov light.

1.2 IDEA detector

An initial electromagnetic calorimeter is proposed in a lead tungstate crystal form, which are readout by SiPMs. It aims for an energy resolution better than $\frac{3\%}{\sqrt{E}} \otimes 1\%$ and includes the possibility of a timing layer.

The magnetic coil is positioned outside the crystal calorimeter and involves a High-Temperature-Superconducting (HTS) solenoid in preliminary development by INFN LASA. The magnetic field produced would be up to 3 T, more than enough depending on the phase of operation of FCC-ee. The high temperature performance (20 – 50K) would also allow a reduction in operational costs and engineering restrictions.

The hadronic calorimeter relies on a fibre based dual-readout calorimeter, making use of alternating rows of scintillating and clear (Cherenkov) fibers embedded in a metal absorber. This design should provide an energy resolution of $\sim \frac{30\%}{\sqrt{E}} \otimes 3\%$. The capillary tubes are oriented with a quasi-pointing geometry towards the interaction vertex, providing high granularity in the transverse plane. This crucial information, together with the crystal calorimeter, provides excellent data for particle flow algorithms.

Muon system

The muon detection system consists of 3 or more layers of detectors, embedded in the iron return yokes that enclose the magnetic field. Its main purpose is to efficiently identify muons and long-lived particles from secondary vertices. The technology used is that of μ -RWELL detectors, a compact gas detector which could provide up to $\sim 100\mu\text{m}$ spatial resolution on the tracks.

Chapter 2

Standard Model

Collider experiments probe the domain of high energy physics, in particular they are mainly understood through the *Standard Model* (SM) of particle physics. The latest formulation (1970s) assumes the categorization of all particles into *elementary particles* and the *mediators* of the forces between them. These two categories are represented by half-integer (*fermions*) and integer (*bosons*) spin values.

The following chapter provides a brief summary of the SM theory involved in the polarization measurements of this work [9] , [10].

2.1 Fermions and bosons

I will give a brief quantitative introduction on the particles that make up the Standard Model; it will then be followed by a summary of the mathematical aspects involved.

Fermions

Fermions, being half-integer spin particles, obey the Pauli exclusion principle, and they are classified further by their carried electric charge into *leptons* and *quarks*. The

2.1 Fermions and bosons

SM defines the electron charge¹ q_e as the elementary electrical charge value; given this convention leptons carry integral electric charge while quarks exhibit fractional charges of $+\frac{2}{3}|q_e|$ or $-\frac{1}{3}|q_e|$. Within leptons and quarks, pairs of particles that exhibit similar behaviour are grouped into generations (*flavors*). This yields 3 generations each: e , μ , τ for leptons and *up/down*, *charm/strange*, *top/bottom* for quarks; these categorizations are summarized in table 2.1.

Particles	Type			$Q/ q_e $
Leptons	e	μ	τ	-1
	ν_e	ν_μ	ν_τ	0
Quarks	u	c	t	$+2/3$
	d	s	b	$-1/3$

Table 2.1: Fermion charges according to the Standard Model.

Particle behaviour, in the quantum mechanical language, is a direct consequence of its mass, electric charge, spin, and other intrinsic quantum numbers. How non-trivial quantum numbers emerge will be treated in more depth in section 2.4.1. In particular, the electric charge is a result of the combination of two more fundamental symmetries: *Weak Isospin* (I) and *Weak Hypercharge* (Y).

Differently from leptons, quarks also exhibit an additional three-fold internal symmetry: *colour*. They also can only exist in bound states with other quarks, a property known as *confinement*.

Bosons

The fundamental interactions between particles in the Standard Model are categorized as: *strong*, *weak*, *electromagnetic* interactions. The *gravitational* interaction is left out of this picture, however, its intensity at the energy scales of particle physics is

¹ $q_e = 1.602\,176\,634 \times 10^{-19} \text{ C}$

negligible. The interactions between SM particles are described as the exchange of bosonic particles, which are responsible for the consequent forces. Their characteristics are summarized as follows:

- Strong interactions are responsible for the bonding of quarks into mesons and barions and their reciprocal interactions. The latter are extremely short ranged, and rapidly decrease to zero above ~ 1 fm. The mediator for strong interactions is the *gluon*, a massless boson which is coupled, with infinite range, only with colour-charged particles (quarks).
- Electromagnetic interactions are coupled with any electrically charged particle. The interaction is expressed through a massless boson: the *photon*.
- Weak interactions are coupled with any fermion and they are mediated by three bosons: W^+ , W^- , Z . These are massive bosons (~ 80 - 90 GeV) and as a consequence they mediate short range interactions.

2.2 The Dirac Equation

The description of fermions is provided by the Dirac equation: it is a relativistically invariant first-order differential equation given by

$$(i\gamma^\mu \partial_\mu - m)\psi = 0 \quad (2.1)$$

where ψ is a four-component spinor and γ^μ are the 4×4 Dirac matrices defined as:

$$\gamma^k = \begin{pmatrix} 0 & \sigma_k \\ -\sigma_k & 0 \end{pmatrix}, \quad \gamma^0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} \quad (2.2)$$

where σ_k are the Pauli matrices. The Dirac equation, while maintaining the energy-momentum relation $E^2 - \mathbf{p}^2 c^2 = m^2 c^4$, implies the existence of negative energy solutions. These are interpreted as a description of *antiparticles* with positive

2.2 The Dirac Equation

energies, namely particles with flipped quantum number signs. The solutions to eq 2.1 describe particle and antiparticle states, each with two possible spin orientations.

It is convenient to work with Pauli-Dirac's representation, that is by decomposing the Dirac 4-vector into 2 bi-dimensional spinors, in analogy to the treatment of spin 1/2 particles:

$$\psi = \begin{pmatrix} \chi \\ \phi \end{pmatrix} \quad (2.3)$$

This basis yields 2 coupled equations:

$$i(\partial_0 + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla})\phi = m\chi \quad (2.4)$$

$$i(\partial_0 - \boldsymbol{\sigma} \cdot \boldsymbol{\nabla})\chi = m\phi \quad (2.5)$$

2.2.1 Helicity and chirality

In the ultra-relativistic limit where the mass term is negligible ($m \rightarrow 0$), the solutions are decoupled into two independent equations describing the evolution of the left-handed and right-handed components separately:

$$i\sigma^\mu \partial_\mu \phi = 0 \quad (2.6)$$

$$i\bar{\sigma}^\mu \partial_\mu \chi = 0 \quad (2.7)$$

When applied to a plane wave solution they produce the **Weyl equations**

$$E\chi = -\boldsymbol{\sigma} \cdot \mathbf{p}\chi \quad (2.8)$$

$$E\phi = +\boldsymbol{\sigma} \cdot \mathbf{p}\phi \quad (2.9)$$

where χ and ϕ are two-component spinors. In total we then find 4 possible solutions, corresponding to two particle/antiparticle states with two possible spin states.

We then introduce the helicity operator $\hat{H}$, which projects the spin onto the direction of momentum:

$$\hat{H} = \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{|\mathbf{p}|} \quad (2.10)$$

Given a massless particle $E = |\mathbf{p}|$, which then implies

$$\hat{H}\chi = -\chi \quad (2.11)$$

$$\hat{H}\phi = +\phi \quad (2.12)$$

Consequently χ and ϕ are eigenstates of the helicity operator, with eigenvalues of helicity ± 1 . It is straightforward to then name these states χ and ϕ as *left-handed* (*LH*) and *right-handed* (*RH*) states.

For a massless particle helicity is then a well defined, Lorentz-invariant quantity; however the description of massive particles requires a different approach. It is clear that for massive particles, since $v \neq c$, helicity is not Lorentz-invariant and therefore not a suitable basis for non ultra-relativistic cases. While helicity depends on the reference frame for massive particles, *chirality* is a Lorentz-invariant property. We can first define the chirality projection operators P_L and P_R for massless particles:

$$P_{L,R} = \frac{1}{2} \left(1 \pm \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E} \right) \quad (2.13)$$

whose action on a generic overlap of LH and RH states ($\Psi = \psi_L + \psi_R$) is therefore:

$$P_L \Psi = \psi_L \quad (2.14)$$

$$P_R \Psi = \psi_R \quad (2.15)$$

In the case of massive fermions, the analogous projection operators act on a 4-component Dirac vector and they are defined as:

$$\hat{P}_{L,R} = \frac{1 \mp \gamma^5}{2} \quad (2.16)$$

where $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ (see eq. 2.2). These operators project the fermion wavefunction on a polarization state $\mathcal{P} = \pm v/c$. We can then incorporate the result from eq. 2.11 as an extreme case where the lepton is ultra-relativistic and is produced in a pure helicity state, meaning $\mathcal{P} \equiv H = \pm 1$.

Chiral states will be the fundamental subjects in the coupling of electroweak currents; however, it seemed appropriate to introduce them in analogy to the simpler helicity states.

2.3 Electroweak theory

2.3 Electroweak theory

A first successful attempt to describe electroweak interaction was made by Enrico Fermi, while describing β decay processes. This attempt, however, was not completely satisfactory: maximal parity violation shifted the theoretical approach to a vector-axial current nature. I will describe the important steps of this process, ending with the current approach adopted by the standard model.

2.3.1 Fermi β decay

In classical electrodynamics the interaction between a charged distribution and the EM field is governed by a dot product $\mathbf{j} \cdot \mathbf{A}$, where j^μ is the current density and A_μ the electromagnetic vector potential. In analogy to what is usually shown with Schrodinger's equation, one can define a conserved probability current for the Dirac equation:² $j^\mu = \bar{\psi}\gamma^\mu\psi$. These ingredients allowed for the first formulation of electroweak processes by Enrico Fermi (1934).

A simple scattering event such as $e + p \rightarrow e + p$ can be modelled by an electro-dynamical interaction between two charged currents with matrix element:

$$M \propto \frac{e^2}{q^2} J_{baryon} J_{lepton} \quad \text{where } \mathbf{q} = \mathbf{p}' - \mathbf{p}$$

In analogy, Fermi postulated that a weak interaction process such as $\nu_e + n \rightarrow p + e^-$ could have a matrix element given by:

$$M = \mathcal{G}_F J_{baryon}^{weak} J_{lepton}^{weak} = \mathcal{G}_F (\bar{\psi}_p \hat{\mathcal{O}} \psi_n) (\bar{\psi}_e \hat{\mathcal{O}} \psi_\nu) \quad (2.17)$$

where $\mathcal{G}_F$ represents Fermi's constant, $\hat{\mathcal{O}}$ a vector operator and $\bar{\psi} = \psi^* \gamma^0$.

It is important to notice the nature of the operator chosen by Fermi: given the success of quantum electrodynamics, it felt natural to introduce an interaction of

²When working with Dirac 4-spinors it is common to introduce the notation of the *adjoint spinor*, such that $\bar{\psi} = \psi^* \gamma^0$

the form $-e\bar{\psi}\gamma^\mu\psi A_\mu$, where the vector current is $\bar{\psi}\gamma^\mu\psi$. Additionally, Fermi was considering parity to be conserved in weak decays, prompting for the choice of a vectorial operator.

2.3.2 V-A theory

A crucial role is played by parity, namely the action of an operator such as $P : f(t, \mathbf{x}) \rightarrow f(t, -\mathbf{x})$. While imposing Lorentz-Invariance, one can construct up to 5 categories of currents under the action of parity:

$$\begin{aligned}\bar{\psi}\psi &= \text{scalar} \\ \bar{\psi}\gamma^5\psi &= \text{pseudoscalar} \\ \bar{\psi}\gamma^\mu\psi &= \text{vector} \\ \bar{\psi}\gamma^\mu\gamma^5\psi &= \text{pseudovector (axial vector)} \\ \bar{\psi}\sigma^{\mu\nu}\psi &= \text{antisymmetric tensor}\end{aligned}$$

Since the operators are acting on 4-component Dirac spinors it is also useful to think of it in these terms: $(1, \gamma^5, \gamma^\mu, \gamma^\mu\gamma^5, \sigma^{\mu\nu})$ constitutes a basis for the space of 4×4 matrices. It is of great importance to notice how axial and vector currents have different behaviours under parity:

$$V : \quad \bar{\psi}\gamma^\mu\psi \quad \xrightarrow{P} \quad \begin{cases} +\bar{\psi}\gamma^0\psi \\ -\bar{\psi}\gamma^k\psi \end{cases} \quad (2.18)$$

$$A : \quad \bar{\psi}\gamma^\mu\gamma^5\psi \quad \xrightarrow{P} \quad \begin{cases} -\bar{\psi}\gamma^0\psi \\ +\bar{\psi}\gamma^k\psi \end{cases} \quad (2.19)$$

While EM and the strong force had been shown to conserve parity, the discovery of parity violation in electroweak interactions (Wu J. 1957) implied a combination of two types of operators with opposite parities. Earlier studies had already shown that in weak decays leptons and antileptons were produced with opposite longitudinal polarisations, narrowing down the choice of interaction to a V or A operators.

2.4 Glashow-Weinberg-Salam model

A generic combination of V and A amplitudes can be described by an operator such as

$$\frac{1}{2}(C_V + \gamma_5 C_A) \quad (2.20)$$

where C_A and C_V are constant coefficients. Neutrinos, however, are produced in pure helicity eigenstates since they are massless (see eq. 2.11); therefore it is required that $C_A = -C_V$, providing then the final form of Fermi's operator

$$\hat{\mathcal{O}} \propto (1 - \gamma_5) \quad (2.21)$$

The operator acts effectively as a projection operator P_L (see eq. 2.16), selecting only the left-handed chiral components of the spinors involved. Consequently, the phenomenological success of the V-A theory implies that the weak interaction is *chiral*: it couples exclusively to left-handed fermions (and right-handed antifermions). Additionally, the theory introduces the concepts of *charged currents* (CC); together with *neutral currents* (NC), they will be the building blocks for the unification of electroweak interactions.

The V-A current is then proportional to a LH projector:

$$J_{V-A}^\mu = \bar{\psi}\gamma^\mu(1 - \gamma^5)\psi \propto \bar{\psi}\gamma^\mu P_L\psi \quad (2.22)$$

and then by using the anticommutation properties of the Dirac matrices ($\{\gamma^\mu, \gamma^5\} = 0$), one can show that $\bar{\psi}\gamma^\mu P_L\psi = \overline{(\psi_L)}\gamma^\mu\psi_L$. This confirms that the $(1 - \gamma^5)$ projects onto the Left-Handed chiral components of the fermion fields to participate in the interaction, consistent with the maximal parity violation observed in the Wu experiment.

2.4 Glashow-Weinberg-Salam model

V-A theory successfully modelled β decays, nevertheless, it was shown to diverge at high energies and a new approach was needed. A final, but crucial, step is to move

to Gauge theory approach; this will naturally produce a field interaction mediated by a massive boson as is currently formulated in the Standard Model.

2.4.1 Gauge invariance

Given the Dirac Lagrangian³ $\mathcal{L} = \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi$ it can be shown that it is invariant for global phase transformation (or global *Gauge invariance*) $\psi \rightarrow e^{i\alpha}\psi$. According to *Noether's theorem*, every continuous global symmetry of the action corresponds to a conserved physical quantity (a charge) and a conserved current J^μ . In the case of Dirac equation this directly describes the conservation of fermion number, namely the number of particles and antiparticles.

By demanding, additionally, $\mathcal{L}$ to be **locally** gauge invariant, namely $\alpha \equiv \alpha(\mathbf{x})$, we need to add terms to the equation and upgrade the standard derivative to a *covariant derivative*. The "new" terms reproduce naturally the electromagnetic potential A^μ and introduce an interaction term with the fermion. This local phase rotation can be introduced as a correspondence to a $\mathbf{U}(1)$ ⁴ symmetry group.

This mechanism provides the template for generating the bosons mediators of the interactions, through the introduction of other symmetry groups.

2.4.2 $\mathbf{SU}(2)_L \otimes \mathbf{U}(1)_Y$

From section 2.3.2 we can conclude that the V-A charged currents are coupled only to left handed fermions (or RH antifermions); we can then organize fermions into LH doublets and RH singlets (or vice-versa), since neutrinos are in a pure and defined helicity state. For example, the first generation:

$$\Psi_L = \begin{pmatrix} \nu_e \\ e^- \end{pmatrix}_L \quad \Psi_R = \begin{pmatrix} e^- \end{pmatrix}_R \quad (2.23)$$

³In a more refined formalism this object is a Lagrangian density per unit volume: $L = \int \mathcal{L} d^4x$

⁴Group of unitary matrices with dimension 1

2.4 Glashow-Weinberg-Salam model

During weak interaction decays the LH components of a same-generation doublet can get mixed. Always mind that leptonic numbers are conserved in any decay. It is therefore natural to introduce the group that mixes LH components while keeping the total probability constant: $\mathbf{SU}(2)_L$ ⁵ or *Weak Isospin* (I).

In analogy to the previous section we can demand the Lagrangian to be invariant under local $SU(2)_L$ transformations. The more complicated non-abelian structure of the group necessitates the introduction of 3 new fields

$$W_\mu^1 \quad W_\mu^2 \quad W_\mu^3 \quad (2.24)$$

together with a coupling constant g . The off-diagonal elements represent the first two charged bosons of weak interactions:

$$W_\mu^\pm = \frac{1}{\sqrt{2}} (W_\mu^1 \mp iW_\mu^2) \quad (2.25)$$

The third field W_μ^3 is a neutral one, it is not however the neutral Z boson yet since it would produce the wrong charges on the Ψ_L doublet.

We now have a $U(1)$ symmetry generating the A_μ potential and the photon field, additionally, by imposing a $SU(2)$ symmetry we found the charged fields for $W^\pm$ bosons and their chiral coupling that generate charged currents. The only piece left to unify electromagnetic and weak interactions is to include neutral currents under the same mathematical approach.

In order to retrieve the natural electric charges (Q) of particles we must introduce another symmetry group, namely $\mathbf{U}(1)_Y$ or *Weak Hypercharge*. Electric charges, in units of fundamental electric charge q_e , can then be defined via the *Gell-Mann-Nishijima* formula:

$$Q = I_3 + \frac{Y}{2} \quad (2.26)$$

The quantum numbers included are the electric charge Q , the third component of isospin I_3 and the corresponding hypercharge Y .

⁵Group of special unitary matrices with dimension 2

Fermion	Label	Chirality	Isospin I	I_3	Charge Q	Hypercharge Y
<i>Leptons</i>						
El. neutrino	ν_{eL}	Left	1/2	+1/2	0	-1
Electron	e_L^-	Left	1/2	-1/2	-1	-1
Electron	e_R^-	Right	0	0	-1	-2
<i>Quarks</i>						
Up quark	u_L	Left	1/2	+1/2	+2/3	+1/3
Down quark	d_L	Left	1/2	-1/2	-1/3	+1/3
Up quark	u_R	Right	0	0	+2/3	+4/3
Down quark	d_R	Right	0	0	-1/3	-2/3

Table 2.2: Weak Isospin and Hypercharge eigenvalues for the first generation of fermions.

To summarize we can apply this to the first generation of fermions: each generation is split into LH and RH chiral pairs. Weak interactions act on LH particle (or RH antiparticle) states, therefore they are a $I = 1/2$ doublet. Meanwhile RH particle (or LH antiparticle) states are not acted upon, therefore they carry no isospin. Hypercharge takes care of retrieving the correct electric charge values. A summary table is reported in table 2.2 .

Finally, the request of local invariance for the $U(1)_Y$ group generates a new field B_μ and a coupling constant g' . The neutral boson we observe as the photon γ and Z^0 are then introduced as orthogonal linear combinations of W_μ^3 and B_μ .

This final formulation predicts the existence of *neutral currents* (NC) mediated by the new Z^0 boson.

2.5 Interaction Lagrangian and Weinberg angle

Given the results obtained in the previous sections we can now define a lorentz invariant lagrangian density for the interaction between charged particles and the bosons. The formulation is in analogy to the electromagnetic interaction mentioned before $q_e \mathbf{J}_\mu^{em} A_\mu$ of an electric current with the EM field.

$$\mathcal{L} = g \mathbf{J}_\mu \cdot \mathbf{W}_\mu + g' J_\mu^Y B_\mu \quad (2.27)$$

Here, $\mathbf{J}_\mu$ and J_μ^Y represent the weak isospin and weak hypercharge currents of the fermions, $\mathbf{W}_\mu$ and B_μ the potentials associated with the boson fields, while g and g' their respective coupling constants.

For comfort we redefine the hypercharge as $Y = Q - I_3$; we can then obtain

$$J_\mu^Y = J_\mu^{em} - J_\mu^{(3)} \quad (2.28)$$

where J_μ^{em} is the electromagnetic current coupled to Q , and $J_\mu^{(3)}$ is the third component of the isospin current.

I shall quickly mention that the steps described until now would generate massless fields; therefore the introduction of bosons masses is done through the Higgs field and spontaneous symmetry breaking. In the case of the γ and Z^0 bosons this can be described by a rotation parametrized by the *Weinberg angle* θ_W :

$$\begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix} = \begin{pmatrix} \cos \theta_W & \sin \theta_W \\ -\sin \theta_W & \cos \theta_W \end{pmatrix} \begin{pmatrix} B_\mu \\ W_\mu^3 \end{pmatrix} \quad (2.29)$$

The Weinberg angle can be defined from the coupling constants

$$\sin \theta_W = \frac{g'}{\sqrt{g^2 + g'^2}} \quad \cos \theta_W = \frac{g}{\sqrt{g^2 + g'^2}} \quad \tan \theta_W = \frac{g'}{g} \quad (2.30)$$

and they yield the inverse forms of $W_\mu^{(3)}$ and B_μ :

$$W_\mu^{(3)} = \frac{g Z_\mu + g' A_\mu}{\sqrt{g^2 + g'^2}} \quad (2.31)$$

$$B_\mu = \frac{-g' Z_\mu + g A_\mu}{\sqrt{g^2 + g'^2}} \quad (2.32)$$

We can now substitute eq. 2.25 , 2.28 , and 2.31 into the Lagrangian 2.27. By naming $J_\mu^\pm = J_\mu^{(1)} \pm iJ_\mu^{(2)}$ we then obtain the final form of the interaction lagrangian:

$$\mathcal{L} = \underbrace{\frac{g}{\sqrt{2}} (J_\mu^- W_\mu^+ + J_\mu^+ W_\mu^-)}_{\text{weak charged current}} + \underbrace{\frac{g}{\cos \theta_W} (J_\mu^{(3)} - \sin^2(\theta_W) J_\mu^{em})}_{\text{weak neutral current}} Z_\mu + \underbrace{g \sin(\theta_W) J_\mu^{em} A_\mu}_{\text{EM neutral current}} \quad (2.33)$$

This equation shows how the interaction energy is composed by terms related to the weak charged and neutral currents, in addition to the already known electromagnetic term. In particular, since we already know the coupling to the EM field A_μ must be the fundamental electric charge we obtain

$$q_e = g \sin \theta_W \quad (2.34)$$

2.5.1 W and Z masses

From the mixing defined in eq. 2.29, provided that the boson described by A_μ is massless and orthogonal to Z^0 , we can obtain a quadratic relationship between the masses:

$$M_Z^2 = M_W^2 \cos^2 \theta_W + M_B^2 \sin^2 \theta_W - 2M_{BW}^2 \cos \theta_W \sin \theta_W \quad (2.35)$$

$$0 = M_\gamma^2 = M_W^2 \sin^2 \theta_W + M_B^2 \cos^2 \theta_W + 2M_{BW}^2 \cos \theta_W \sin \theta_W \quad (2.36)$$

$$0 = M_{Z\gamma}^2 = (M_W^2 - M_B^2) \sin \theta_W \cos \theta_W + M_{BW}^2 (\cos^2 \theta_W - \sin^2 \theta_W) \quad (2.37)$$

By dropping off-diagonal terms we can obtain an experimental relationship for Weinberg's angle involved in mass couplings of the weak bosons [11]:

$$\sin^2 \theta_W = 1 - \left(\frac{M_W}{M_Z} \right)^2 = 0.23129 \pm 0.00004 \quad (2.38)$$

2.5.2 Leptons coupling with Z

In this study we will make use of the couplings of fermions with the neutral boson Z^0 . The interaction Lagrangian (eq. 2.33) already shows that the fermion coupling to

2.5 Interaction Lagrangian and Weinberg angle

Lepton	Q	C_A	C_V
ν_e, ν_μ, ν_τ	0	$+\frac{1}{2}$	$+\frac{1}{2}$
e, μ, τ	-1	$-\frac{1}{2}$	$-\frac{1}{2} + 2 \sin^2 \theta_W$

Table 2.3: Axial and Vector coupling of leptons with the neutral boson Z .

the neutral current is $\propto J_\mu^{(3)} - \sin^2(\theta_W) J_\mu^{em}$. Given the LH and RH states described as isospin doublets (see eq. 2.23 and table 2.2), it is useful to introduce coupling constants for LH and RH fermions:

$$g_L = I_3 - Q \sin^2 \theta_W \quad g_R = -Q \sin^2 \theta_W \quad (2.39)$$

where Q is the electric charge and I_3 the third component of weak isospin. Right handed fermion coupling, indeed, has $I_3 = 0$ since it's not included by the chiral nature of weak interactions.

The vector and axial components of the interaction couple LH and RH states with same and opposite sign, respectively. Therefore the V and A coefficients are as follows:

$$C_V = g_L + g_R = I_3 - 2Q \sin^2 \theta_W \quad C_A = g_L - g_R = I_3 \quad (2.40)$$

The axial and vector couplings for leptons are reported in table 2.3.

Chapter 3

Z production and τ polarization

This chapter focuses on describing the conditions at the Z resonance and how the polarization estimates can be obtained. I will also detail the necessary kinematic distributions that will be used when fitting the data in my analysis.

3.1 Z resonance

As introduced in chapter 1, FCC-ee is expected to have an operational phase dedicated to the production of Z bosons, with 6×10^{12} expected Z decays. The most interesting process, for the purpose of this work, is $e^+e^- \rightarrow Z/\gamma \rightarrow f\bar{f}$ where an electron and a positron collide and decay into a fermion and anti-fermion pair. This process is governed by the exchange of a photon or the Z boson, and its cross section can be generally written as:

$$\frac{d\sigma}{d\Omega} = \left(\frac{d\sigma}{d\Omega}\right)_{\text{QED}} + \left(\frac{d\sigma}{d\Omega}\right)_{\text{weak}} + \left(\frac{d\sigma}{d\Omega}\right)_{\text{interference}} \quad (3.1)$$

The first contribution (QED) comes from the photon exchange, the second one (weak) from the Z exchange and the third is due to their interference. While the contributions are experimentally undistinguishable, the cross section exhibits an

3.2 Forward-Backward and Polarization asymmetries

ordinary $1/s$ fall-off when moving from the Z pole energy $\sqrt{s} = m_Z$ [2], as can be seen in figure 3.1, while around the Z pole it has an enhancement of the order m_Z^2 .

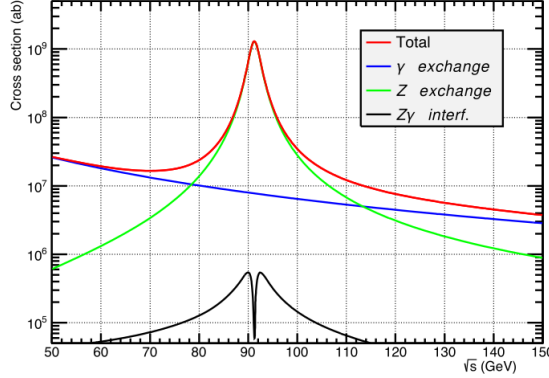


Figure 3.1: Contributions to the e^+e^- cross section as a function of centre of mass energy $\sqrt{s}$ [12]. The distinct peak at the resonance mass illustrates the Z-pole mechanism to generate the expected yield at FCC-ee.

The photon exchange is therefore dominated by the Z interaction at energies corresponding to the Z pole. Apart from the energy dependence of some observables due to interference, the contribution of photons is then negligible at energies $\sqrt{s} = m_Z$. This peculiarity allows FCC-ee to serve its role as a Z factory and to achieve the expected luminosities when operating at the Z pole.

3.2 Forward-Backward and Polarization asymmetries

The $e^+e^- \rightarrow f\bar{f}$ process, being dominated by the Z interaction at $\sqrt{s} = m_Z$, naturally manifests parity violation being an electro-weak interaction. The final states of this process will therefore exhibit natural asymmetries that arise from said violation.

Among the opportunities for experimental physics at the Z pole, *forward-backwards* and *polarization* asymmetries are powerful tools to probe the value of $\sin \theta_W^{eff}$.

Forward-Backward

The Forward-Backward asymmetry quantifies the tendency of the produced (anti)fermions to be in the emitted in the directions of the incoming (anti)electron beam, versus the opposite direction. Parity violation causes this distribution to be asymmetric across the two hemispheres. Experimentally the asymmetry can be defined by the cross sections for forward (F) and backwards (B) hemispheres:

$$A_{FB} = \frac{\sigma_F - \sigma_B}{\sigma_F + \sigma_B} \quad (3.2)$$

Recall from section 2.5.2 that fermions coupling to the Z boson is expressed through the axial and vectorial constants g_V and g_A ¹. We can define an asymmetry parameter $\mathcal{A}_f$ to express quantify how much the coupling differs from a pure V-A interaction for a given fermion f :

$$\mathcal{A}_f = 2 \frac{g_{V_f}/g_{A_f}}{1 + (g_{V_f}/g_{A_f})^2} \quad (3.3)$$

In the $f\bar{f}$ process at the Z pole energy the Forward-Backwards asymmetry is directly proportional to the asymmetry parameters for the leptons involved:

$$A_{FB} \equiv A_{FB}^0 = \frac{3}{4} \mathcal{A}_e^0 \mathcal{A}_f^0 \quad (3.4)$$

Polarization

While e^+e^- beams are expected to be depolarized at FCC-ee, parity violation is such that in the case of $f\bar{f} = \tau^+\tau^-$ a non-zero polarization is expected. The polarization

¹These were previously introduced as C_V and C_A , in agreement with a more traditional notation; they are now redefined $g_{A,V}$.

3.2 Forward-Backward and Polarization asymmetries

$\mathcal{P}_\tau$ is defined as:

$$\mathcal{P}_\tau = \frac{\sigma_R - \sigma_L}{\sigma_R + \sigma_L} \quad (3.5)$$

where σ_R and σ_L are the cross-sections for the production of right-handed and left-handed τ^- leptons, respectively.

The polarization is also dependent on the angle θ , defined as the angle between the incoming e^- and the outgoing τ^- . The following expression allows for the estimation of the electron and tau couplings, $\mathcal{A}_e$ and $\mathcal{A}_\tau$:

$$\mathcal{P}_\tau(\cos \theta) = -\frac{\mathcal{A}_\tau(1 + \cos^2 \theta) + 2\mathcal{A}_e \cos \theta}{(1 + \cos^2 \theta) + 2\mathcal{A}_e \mathcal{A}_\tau \cos \theta} \quad (3.6)$$

The mean polarization can then be obtained by integrating over all possible angles and yields a direct relationship with the τ coupling:

$$\langle \mathcal{P}_\tau \rangle = -\mathcal{A}_\tau \quad (3.7)$$

A crucial feature of the polarization measurement is the ability to extract $\mathcal{A}_e$ and $\mathcal{A}_\tau$ independently. While the Standard Model assumes *Lepton Universality*, implying $\mathcal{A}_e = \mathcal{A}_\tau = \mathcal{A}_\mu$, this measurement provides the opportunity to fit for both onto an angular distribution. These methods enable the experimental measurements of the asymmetry parameters and therefore an indirect estimate of the parameter $\sin \theta_W^{eff}$, simply by re-arranging equations 3.3 and 2.40 :

$$\frac{g_V}{g_A} = 1 - 4|Q_f| \sin^2 \theta_W^{eff} \quad (3.8)$$

3.2.1 Polarization and kinematics

This study focuses only on measurements of the polarization asymmetry and its techniques. It is important to notice that experimentally we have access to a **helicity** measurement, while technically the weak-current are defined on **chiral states**. The running conditions of FCC-ee and previous accelerators, however, are such that the

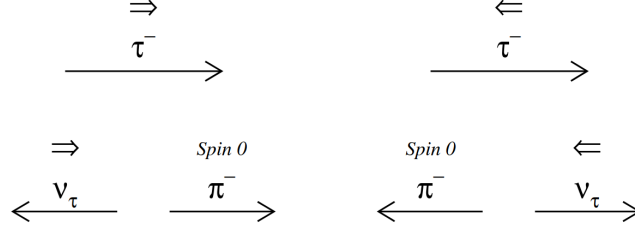


Figure 3.2: Schematic overview of the two possible helicity states of the decay mode $\tau^- \rightarrow \pi^- \nu_\tau$. The long arrow $\longrightarrow$ indicates the particle momentum, while the short double arrow $\Rightarrow$ indicates the particle spin. [13]

particles are produced in highly relativistic states which are identical to helicity states, as shown in section 2.2.1. The polarization is obtained by fitting channel-specific kinematic observables, which are dependent on the original tau polarization. Both mean and angular-dependent polarization are tested in this study; they both rely on the following definitions of optimal kinematic variable for each decay channel.

In order to understand how polarization effects the kinematics a simple example can be shown [13]. Consider a weak decay $\tau \rightarrow \pi \nu_\tau$: since neutrinos are massless they are in a pure helicity state $h = \pm 1$ and therefore, thanks to the conservation of angular momenta, the pion produced will be preferably aligned with the original spin of the tau. Pions are in fact a spin 0 particle, and therefore the spin information is carried only by the neutrino. This means that on average, in the laboratory frame, a pion produced by a $h = +1$ tau will be more energetic than one produced by a $h = -1$ tau. A simple scheme can be seen in figure 3.2. The opposite can then be said for a positive tau decay $\tau^+ \rightarrow \pi^+ \bar{\nu}_\tau$ (recall that the helicity is such that $\nu = \nu_L$ and $\bar{\nu} = \bar{\nu}_R$). Since the $\tau^+ \tau^-$ produced by the Z^0 decay have opposite helicities, given the same (negative) $\mathcal{P}_\tau \equiv \mathcal{P}_{\tau^-} \equiv -\mathcal{P}_{\tau^+}$ polarization, the weak boson is expected to favour $\tau_L^- \tau_R^+$ configurations.

A similar description can be made of the rho decay $\tau \rightarrow \rho \nu_\tau \rightarrow \pi \pi^0 \nu_\tau$, with

3.2 Forward-Backward and Polarization asymmetries

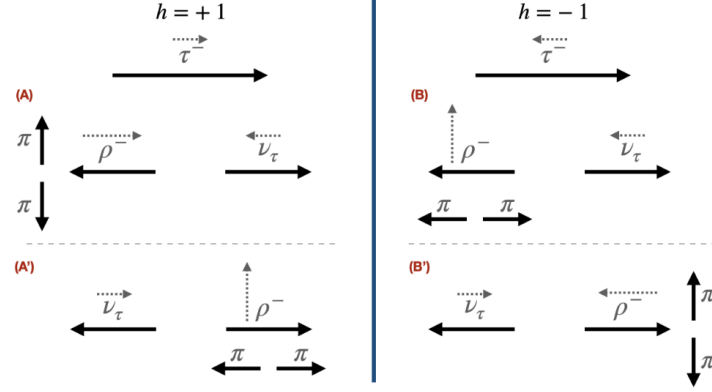


Figure 3.3: Schematic overview of the decay possibilities of a positive and negative helicity τ^- in a hadronic ρ decay. The dashed lines represent spin orientation.

the additional complication that the ρ is a spin 1 particle, and therefore the spin information is divided into the neutrino and the hadron. Conservation of angular momenta yields the two possibilities that the ρ spin is either aligned or perpendicular to the tau spin. Consider the schematic view in figure 3.3, for example the decay of a positive helicity τ^- :

- Configuration **A**: the neutrino has to maintain $S_{\nu,z} = -\frac{1}{2}$ given its helicity, while the ρ hadron is forced into a spin $S_{\rho,z} = +1$ state.
- Configuration **A'**: the neutrino has to maintain $S_{\nu,z} = +\frac{1}{2}$ given its helicity, and forces the rho into a spin $S_{\rho,z} = 0$ state.

The ρ hadrons produced in case A are then on average less energetic than the A' configuration. This means that both the helicity configurations can yield decays that cover most of the energetic spectrum. A similar description can be made for leptonic decays $\tau \rightarrow e\nu_e\nu_\tau$ or $\tau \rightarrow \mu\nu_\mu\nu_\tau$. This overlap of hard and soft hadrons necessitates the introduction of different kinematic variables to recover the full sensitivity to the polarization, in contrast with the pion channel.

3.3 Optimal observables

This study focuses on the main decay channels, in order to cover most of the branching ratios of the τ lepton; a full table is shown below at 5.1.

Type	Decay Mode	Resonance	BR [%]
<i>Leptonic Decays</i>			
	$\tau^- \rightarrow e^- \bar{\nu}_e \nu_\tau$	-	17.82 ± 0.04
	$\tau^- \rightarrow \mu^- \bar{\nu}_\mu \nu_\tau$	-	17.39 ± 0.04
<i>Hadronic Decays (1-prong)</i>			
	$\tau^- \rightarrow \pi^- \nu_\tau$	π^-	10.82 ± 0.05
	$\tau^- \rightarrow \pi^- \pi^0 \nu_\tau$	ρ^-	25.49 ± 0.09
	$\tau^- \rightarrow \pi^- 2\pi^0 \nu_\tau$	a_1^-	9.26 ± 0.10
<i>Hadronic Decays (3-prong)</i>			
	$\tau^- \rightarrow \pi^- \pi^+ \pi^- \nu_\tau$	a_1^-	8.99 ± 0.05
Total			~ 90

Table 3.1: Analysed τ decay modes, covering around $\sim 90\%$ of the branching ratios [11].

Each decay follows a specific distribution which depends on the polarization, and therefore allows to perform a fit over the reconstructed data to extract the mean tau polarization. The experimentally accessible kinematic quantities are the momenta $\mathbf{p}$ of the hadrons and charged leptons, and the tau energy $\sqrt{s}/2$. It is not then possible to work in the τ rest frame and all the polarization estimators must be defined in the laboratory frame [14].

Given the set of all n kinematic observables ξ , the decay distribution can always be written as the sum of polarization dependent and independent parts [15]:

$$W(\xi) = f(\xi) + \mathcal{P}_\tau g(\xi) \quad (3.9)$$

3.3 Optimal observables

These functions satisfy the following normalization conditions:

$$\int f(\xi) d^n \xi = 1 \quad , \quad \int g(\xi) d^n \xi = 0 \quad , \quad f \geq 0 \quad , \quad |g| \leq f \quad (3.10)$$

The condition $\int g d\xi = 0$ expresses the fact that the total decay width does not depend from the polarization. In addition, $W(\xi)$ is proportional to the squared matrix element, and therefore must be defined positive via $f \geq 0$ and $|g| \leq f$. We then define the **optimal variable** ω as:

$$\omega = \frac{g(\xi)}{f(\xi)} = \frac{W^+(\xi) - W^-(\xi)}{W^+(\xi) + W^-(\xi)} \quad \text{with domain} \quad \omega \in [-1, +1] \quad (3.11)$$

where $W^\pm$ are the decay distribution for positive and negative helicity.

The method of *maximum likelihood* allows to demonstrate that the optimal variable measures the polarization with the maximum sensitivity allowed. Given $\{\xi_i\}$ the set of N measurements, the likelihood function is defined as:

$$\mathcal{L}(\xi_i, \mathcal{P}_\tau) = \prod_{i=0}^N (f(\xi_i) + \mathcal{P}_\tau g(\xi_i)) \quad (3.12)$$

The measured $\mathcal{P}_\tau^{\text{exp}}$ is the value that maximizes the likelihood, or in simpler terms its logarithm:

$$\frac{\partial \log \mathcal{L}}{\partial \mathcal{P}_\tau^{\text{exp}}} = \sum_j \frac{g(\xi_j)}{f(\xi_j) + \mathcal{P}_\tau^{\text{exp}} g(\xi_j)} = \sum_j \frac{\omega_j}{1 + \mathcal{P}_\tau^{\text{exp}} \omega_j} = 0 \quad (3.13)$$

The error on the polarization measurement is then given by:

$$\frac{1}{\sigma_{\mathcal{P}_\tau}^2} = -\frac{\partial^2 \log \mathcal{L}}{\partial \mathcal{P}_\tau^2} = N \left\langle \left(\frac{\omega}{1 + \mathcal{P}_\tau \omega} \right)^2 \right\rangle \quad (3.14)$$

which yields a per-channel sensitivity of

$$S = \frac{1}{\sigma_{\mathcal{P}_\tau} \sqrt{N}} = \sqrt{\left\langle \left(\frac{\omega}{1 + \mathcal{P}_\tau \omega} \right)^2 \right\rangle} \quad (3.15)$$

3.3.1 Leptonic channels

Neglecting mass and radiative effects, the decay distribution for the electron and muon is essentially identical. By defining the kinematic variable in the laboratory frame

$$x = \frac{E_{lepton}}{E_\tau} \quad (3.16)$$

the leptonic decay distribution can be derived as:

$$W_{lep}(x) = \frac{1}{3}[a(x) + \mathcal{P}_\tau b(x)] = \frac{1}{3}[(5 - 9x^2 + 4x^3) + \mathcal{P}_\tau(1 - 9x^2 + 8x^3)] \quad (3.17)$$

The suitable optimal variable is therefore defined as $\omega = b(x)/a(x)$; nonetheless for leptonic decays it is shown to be equivalent to perform a fit over the optimal variable or the decay distribution $W(x)$.

3.3.2 Hadronic channels

In analogy, for the pion channel the only observable quantity is the energy of the charged hadron. The variable is defined as the lepton channel $x_\pi = E_\pi/E_\tau$ and its decay distribution is shown to be

$$W(x_\pi) = 1 + \mathcal{P}_\tau(2x_\pi - 1) \quad (3.18)$$

Multipion decays, on the other hand, are influenced by multiple kinematic parameters given their more complex decays. Any hadronic decay distribution can be written in the generic form of [14]

$$\frac{dN}{d\cos\theta^*} = \frac{1}{2}(1 + \alpha\mathcal{P}_\tau\cos\theta^*) \quad \alpha = \frac{m_\tau^2 - 2m^2}{m_\tau^2 + 2m^2} \quad (3.19)$$

where θ^* is the decay angle between the hadron and the tau, in the tau rest frame, and m is the invariant mass of the hadronic system. The proportionality factor yields for ρ and a_1 respectively: $\alpha_\rho \sim 0.46$ and $\alpha_{a_1} \sim 0.12$, that reflects a reduced sensitivity with respect to pion decays.

3.3 Optimal observables

Some sensitivity can be regained by spin-analysing the hadronic decay itself. In order to do so, the optimal decay variable has to be introduced.

Rho decays

The decay distribution for positive and negative helicity ρ decay is defined as follows [16]:

$$W^\pm = \frac{3}{4(m_\tau^2 + 2m^2)} \left[w_1^\pm(m, \psi_\tau) h_1(\psi_\rho) + w_0^\pm(m, \psi_\tau) h_0(\psi_\rho) + w_{-1}^\pm(m, \psi_\tau) h_{-1}(\psi_\rho) \right] \quad (3.20)$$

where the τ decay angle ψ_τ is given by the total hadronic energy E_h and the hadronic invariant mass m

$$\cos \psi_\tau = \frac{2E_h/E_{\text{beam}} - 1 - m^2/m_\tau^2}{1 - m^2/m_\tau^2} \quad (3.21)$$

The functions $w_0^\pm, w_1^\pm, w_{-1}^\pm$, to be distinguished from the optimal variable ω , are elements of the hadron density matrix and they are given by:

$$w_0^+ = \left(m_\tau \cos \eta \cos \frac{\psi_\tau}{2} + m \sin \eta \sin \frac{\psi_\tau}{2} \right)^2 \quad (3.22)$$

$$w_0^- = \left(m_\tau \cos \eta \sin \frac{\psi_\tau}{2} - m \sin \eta \cos \frac{\psi_\tau}{2} \right)^2 \quad (3.23)$$

$$w_1^+ + w_{-1}^+ = \left(m_\tau \sin \eta \cos \frac{\psi_\tau}{2} - m \cos \eta \sin \frac{\psi_\tau}{2} \right)^2 + m^2 \sin^2 \frac{\psi_\tau}{2} \quad (3.24)$$

$$w_1^- + w_{-1}^- = \left(m_\tau \sin \eta \sin \frac{\psi_\tau}{2} + m \cos \eta \cos \frac{\psi_\tau}{2} \right)^2 + m^2 \cos^2 \frac{\psi_\tau}{2} \quad (3.25)$$

The components of the spin 1 state have to be rotated by a Wigner rotation term η which, by neglecting higher terms $\mathcal{O}(m_\tau^2/M_Z^2)$, can be obtained via :

$$\tan \frac{\eta}{2} = \frac{m}{m_\tau} \tan \frac{\psi_\tau}{2} \quad (3.26)$$

The only experimentally accessible observable for the ρ decay is the angle ψ_ρ between the direction of the charged π and the rho's flight in its rest frame. It can be expressed

with kinematic variables as [14]:

$$\cos \psi_\rho = \frac{m}{\sqrt{m^2 - 4m_\tau^2}} \frac{E_1 - E_2}{|\mathbf{p}_1 + \mathbf{p}_2|} \quad (3.27)$$

with $\mathbf{p}_i$ and E_i of the two pions $\pi^\pm \pi^0$ produce by the rho decay. Finally, the h functions are defined as:

$$h_1 = h_{-1} = \sin^2 \psi_\rho \quad (3.28)$$

$$h_0 = 2 \cos^2 \psi_\rho \quad (3.29)$$

After gathering all the necessary information, the optimal variable ω can be constructed by substituting the decay distributions 3.20 in the definition 3.11.

Expected sensitivity

As briefly mentioned before, the optimal variables allow to recover most of the sensitivity in the polarization measurement. Additional information can be obtained by considering all kinematic observables, especially for a_1 decays. The last available information is the τ flight direction $\hat{\tau}$; while this allows for the full recovery of sensitivity it is a very complicated task, due to the presence of final neutrinos and other limitations of the study. A comprehensive table of the expected sensitivities and recoveries is shown below in table 3.2.

Given the complexity of the a_1 decay and the need to use more complicated kinematic angles to retrieve sensitivity, it was excluded from the polarization estimates.

3.3 Optimal observables

Number of observables				
Decay	1	2	all but $\hat{\tau}$	with $\hat{\tau}$
$l\nu\bar{\nu}$	0.22	-	0.22	0.27
$\pi\nu$	0.58	-	0.58	0.58
$\rho\nu$	0.26	0.49	0.49	0.58
$a_1\nu$	0.10	0.23	0.45	0.58

Table 3.2: Expected theoretical sensitivities, according to the number of kinematic observables used. Sensitivities closer to 1 yield a more precise measurement. The columns represent increasing levels of kinematic information: only using the visible energy, adding the two internal decay angles, an additional azimuthal angle for a_1 decays, and finally including the $\hat{\tau}$ flight direction.

Chapter 4

Data analysis

In this chapter I will provide an overview of the analysis performed; I will also show the performance of the event identification applied and the fitting techniques. The complete analysis code is available on my personal public Github repository at:

<https://github.com/MicheleScattola/FCCAnalyses.git>

The datasets of this analysis are available on CERN's EOS at the following path:

`/eos/experiment/fcc/ee/generation/DelphesEvents/
/winter2023/IDEA/p8_ee_Ztautau_ecm91/`

4.1 DELPHES

The events used in this analysis are a product of complex simulations provided by tools such as PYTHIA8 and DELPHES. The first one is a general purpose Monte Carlo program; it makes use of detailed physical models in order to provide high-energy physics collision events. The latter one, DELPHES; is a C++ framework built to perform a fast multipurpose detector response simulation. Monte Carlo events generated by PYTHIA8 are loaded into a DELPHES rendition of the IDEA detector; the output obtained is then representative of the expected performance. These

4.1 DELPHES

programs are progressively being included into a robust and widespread environment called FCCAnalyses, which is built to develop code for the FCC Project. While a complete description of the simulation is outside of the scope of this thesis, it is useful to summarize how the parametric response of the detector is obtained by DELPHES.

4.1.1 Detector simulation

Given the set of track parameters $\vec{\alpha}$, the track trajectory is a function of said parameters $\vec{x} = \vec{x}(\vec{\alpha})$. The passage of the track through the detectors is simulated by estimating the covariance matrix C of the track parameters; this is done through the use of a Kalman method:

- The detector is implemented in a simplified geometrical layout in order to maintain fast execution times.
- Starting from the most outer layer, the trajectory is tracked through the detector and for each measurement layer the covariance matrix is updated. First, the expected detector resolution is added to diagonal elements of C , then the off-diagonal contributions are added to represent multiple scattering events.
- At the end of this process the tracks parameters are randomly smeared in agreement with the covariance matrix, such that $\vec{\alpha}_{\text{reconstructed}} = \vec{\alpha}_{\text{true}} + \delta\vec{\alpha}$

This process produces randomly generated values for track parameters, according to the expected detector performances, which are then the simulated reconstructed values. The DELPHES framework also generates simplified representation of showers in the calorimeter and hits in the muon chambers. These data are then fed to a *Particle Flow* algorithm that recognises basic particle groups, depending on their characteristic traces left in the passage through the detector. These categories include *electrons* , *muons* , *photons* and *charged hadrons*.

It is important to report that depending on the generation of the data simulation the detector can have different recognition performances. The detector version is indeed embedded into a *card* configuration file, and depending on the chosen card for the data generation the particle flow can have certain efficiencies, especially for particle recognition at lower energy scales. This will result in evident detector performance limitation, shown in section 4.2.3.

4.1.2 Data output

The simulated reconstructed data are then stored, together with Monte Carlo data, in the output file. The extension used is a ROOT file, structured in the format of Event Data Model for High Energy Physics (EDM4hep). Fundamentally, it is a flat tree structure for fast data access: a principal tree called "events" stores all events simulated and each event is accessed separately. For each event the data are organized into collections such as:

- **ReconstructedParticles** contains all simulated values which play the role of reconstructed particles in the detector. For the sake of brevity, this collection will be referenced as "RECO".
- **Particle** contains all Monte Carlo particles , which represent the particles before detector effects.
- The Particle Flow algorithm provides collections for **Muons** , **Electrons**, **Photons** ; these are links that reference items in the RECO collection.

Each particle item in the collections is an element of a pre-defined struct which holds the actual numerical values of some key parameters. The Monte Carlo particles are stored as a `edm4hep::MCParticleData` class element, while RECO particles as `edm4hep::ReconstructedParticleData`. These classes share some typical variables such as momentum $\vec{p}$, mass and charge. A schematic summary is visibile in figure 4.1 .

4.1 DELPHES

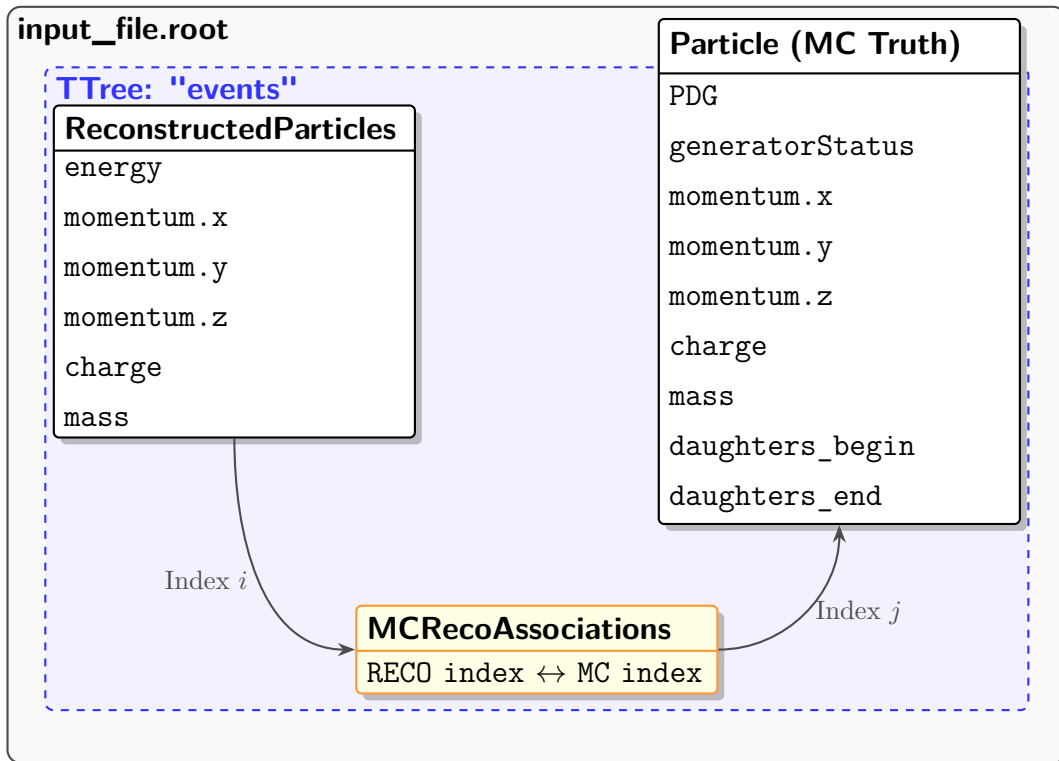


Figure 4.1: Schematic summary of the EDM4hep file structure.

RECO particles have their mass variable assigned by the Particle Flow algorithm which decides which particle is most likely to be represented by that track. In particular, charged hadrons have similar shower profiles in the calorimeter and therefore they are generically all assigned the charged pion mass (m_π); additional information, however, might change the Particle-ID mass to that of Kaons or protons. In this study pions have been selected in this exact way, by filtering reconstructed particles with those that had $m = m_\pi$.

MC particles have additional properties such as:

- **PDG** : Particle Data Group ID, it's a unique particle id that represents a type of particle (eg. $\mu^- = 13$, $\mu^+ = -13$, $\pi^0 = 111$) .
- **generatorStatus** : an integer representing the status of the Monte Carlo particle inside the generator process. In the EDM4hep format the reported statuses are: 4 for beam particles, 2 for decayed or intermediate particles and 1 for final stable particles. This will be extremely helpful in identifying the decayed particles, especially to navigate through the Monte Carlo decays.
- **daughters_begin / daughters_end** : these are integer indices in the ordered Particle collection. Monte Carlo particles are, in fact, ordered in sequence following the decay chain. Given a particle **A** which is element **Particle[a]** in the Particle block, we can find all of its daughter particles in the elements at indices:

`A.daughters_begin < index < A.daughters_end`

`Daughters = { Particle[A.daughters_begin], Particle[A.daughters_begin + 1],
 , ... , Particle[A.daughters_end - 1] }`

An example of event chain with particle indices and genStatus is reported in figure 4.2.

4.2 Event selection

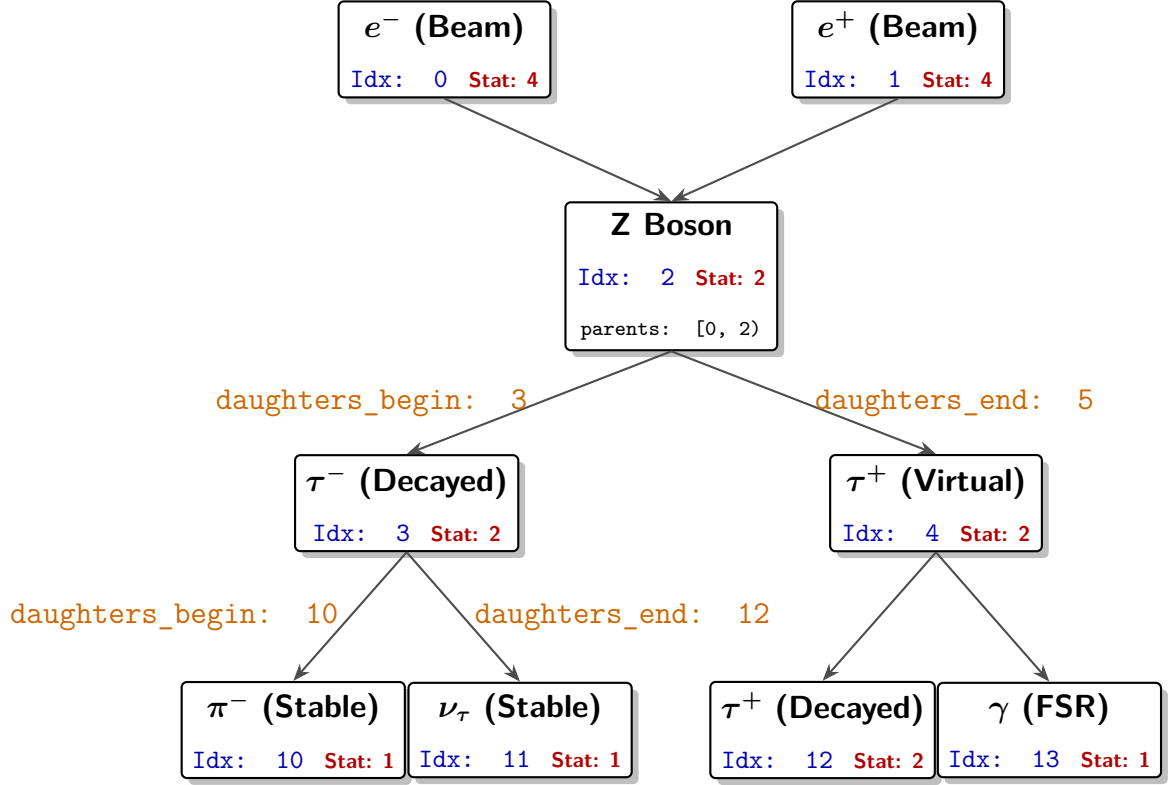


Figure 4.2: Example of an $ee \rightarrow Z \rightarrow \tau\tau$ process: Particle indices are progressive and status indicate stable or decayed particles. As it often happens one of the τ can undergo Final State Radiation (FSR) before stabilizing.

4.2 Event selection

The analysis is performed by accessing the `.root` data files and constructing the necessary ROOT Dataframe in order to extract the reconstructed variables. The code used has mostly been developed in C++ and `python` and makes ample use of ROOT libraries and objects.

I will now give a detailed description of the event selection process developed, following the logical order that is implemented in the code. A more detailed description

with direct references to the `C++` functions can be found in the `README.md` file on the git repository linked at the beginning of this chapter.

4.2.1 Pre-selection

The $Z \rightarrow \tau\tau$ decay's events are characterized by two back-to-back jets; I then needed establish the axis of the decay, namely the *thrust vector*. It is defined as the vector that maximizes the projection of the visible particles momentums:

$$T \equiv \max_{\hat{\mathbf{n}}} \frac{\sum_i |\mathbf{p}_i \cdot \hat{\mathbf{n}}|}{\sum_i |\mathbf{p}_i|} \quad (4.1)$$

The thrust vector is calculated by looping over all reconstructed particles and calling a pre-existing algorithm from `FCCAnalyses::Algorithms::minimize_thrust`. In order to do so, the first constraint is introduced, which is to require at least 2 reconstructed particles. By defining the thrust axis, the event can be divided into the two back-to-back hemispheres; the reconstructed particles are then assigned a hemisphere based on the $\cos \theta$ between their momentum and thrust vector.

Given the subdivision into hemispheres, the total reconstructed charge in each hemisphere is calculated. To further reduce bias, the direction of thrust is always taken from the negatively charged hemisphere to the positively charged. This introduces a second constraint, which is to have opposite non-null charges reconstructed in the two hemispheres. Note that this requirement is also consistent with the decay of a neutral boson such as the Z^0 .

Lastly, in order to ensure that the event is fully contained in the detector stereo-angle coverage, a $|\cos \theta_{\text{thrust}}| < 0.95$ limit is applied.

These requirements yield a pre-selection efficiency of around $\sim 92\%$ on the events considered.

4.2 Event selection

4.2.2 Event identification

Once the pre-selection was applied I moved onto identifying the reconstructed particles and implementing a classification for the event. An event classification was also applied for Monte Carlo particles.

Monte Carlo particles

The basic idea for Monte Carlo identification is to find the decayed τ in the Particle collection, and then navigate through its daughters and recognize particle types by their PDG property. The decayed τ MC particles are identified by looping through the Particle branch and imposing certain conditions. For each hemisphere the following process is applied:

- `abs(PDG) == 15` : τ particles are identified by their PDG integer of ± 15 . The particle/antiparticle sign is checked by the particle charge.
- `$\mathbf{p}_\tau \cdot \mathbf{p}_{TOT}^{RECO} > 0$` : this ensures that the Monte Carlo tau is oriented towards the same hemisphere as the reconstructed particles, in the same hemisphere. Note that an equivalent condition would be to impose the same charge as the reconstructed particle. A kinematic condition was preferred since there could be edge cases with badly reconstructed charge. Since opposite well-reconstructed charges are imposed in the pre-selection, this kinematic approach is effectively the same condition, but it's safer for a generic analysis.
- `generatorStatus == 2` : Sometimes in the PYTHIA8 rendition of a decay, if the particle goes through some radiative process before decaying, there can be multiple copies of the particle before and after the radiation emission. Imposing the generator status filters by collecting only particles that are actually decayed. Additionally a safety check is performed by requiring that the particle does not have any τ daughters; this would mean that the decay is not the final one.

At the end of this process a certain Monte Carlo particle is identified as the correct τ of that hemisphere's decay. By navigating through its daughter particles I collect all of their PDGs and therefore identify the decay chain of that particular event.

Reconstructed Particles

As previously introduced, some particles are recognized by Particle Flow algorithms and therefore they are pre-classified in specific branches in the root file. Available branches for reconstructed particles in this analysis are: muons, electrons, photons. Pions, on the other hand, did not have a specific branch but the most straightforward identification is by filtering RECO particles with mass values attributed as pions. This is because pions, contrary to the other branches, are not reconstructed with such high degrees of confidence and can be mistaken for Kaons or other hadrons. That being said, the data samples used did show some form of distinction between Kaon and pi traces but they are not in general as reliable as the identification of electrons, muons and photons.

Each of these identified collections is filtered by imposing the correct hemisphere based on the thrust axis, as described in the section above. From this process we get collections of μ , e , γ and $\pi^\pm$ filled with the corresponding particles which have been reconstructed in that hemisphere. The identification process I implemented required simply to count the number of elements for each collection.

For leptonic decays the choice is simple, 1 lepton and 0 charged pions identify the decay. Hadronic decays require more attention since they are heavily influenced by the decay $\pi^0 \rightarrow \gamma\gamma$. Photons can either be lost, in the case of very low energy, or even merged into one single calorimetric trace in cases of extreme collinearity. A conservative and simple approach was then implemented in order to reduce cross-channel background. The main overlaps can be found in ρ decays being reconstructed as single π , or a_1 decays interchangeably miss-identified as ρ . This is because, as

4.2 Event selection

stated, some photons can be lost or miss-counted in the resonant decays

$$\rho \rightarrow \pi\pi^0 \rightarrow \pi\gamma\gamma$$

$$a_1 \rightarrow \pi\pi^0\pi^0 \rightarrow \pi\gamma\gamma\gamma\gamma$$

The decided requirements are as follows:

- Pion: 1 reconstructed π and 0 γ .
- Rho: 1 reconstructed π and 1 or 2 γ .
- a_1 (1-prong): 1 reconstructed π and 3 or more γ .
- a_1 (3-prong): 3 reconstructed π .

A final filter is applied at the total visible invariant mass: while in theory all events should respect $m_{\text{vis}} \leq m_\tau \sim 1.77 \text{ GeV}$ it is clear that a tight limit might be too restrictive for events whose mass is already near the threshold, for example a_1 3-prong decays. A more permissive limit is imposed of $m_{\text{visible}} < 2 \text{ GeV}$, in order to reject events which are blatantly miss-reconstructed.

Resonant decays could also be filtered by their reconstructed mass; in figure 4.3 invariant masses for ρ and a_1 channels are shown. While introducing a mass-window to identify the channel may sound helpful, it is not straightforward that they might be efficient. I tested two mass-windows identified by looking at the invariant mass distributions from the reconstructed event; the tests I performed were over a selection defined by:

$$0.3 \text{ GeV} < m_\rho < 1.2 \text{ GeV} \quad 0.6 \text{ GeV} < m_{a_1} < 1.8 \text{ GeV}$$

In order to check their efficiency I analysed the rejection rate of background events; full results are shown in table 4.1. The filter was shown to be effective for both 1 and 2 photons rho events: respectively $\sim 41\%$ and $\sim 60\%$ of the rejected events were correctly excluded. This means that if I were to include them, those events

would yield only $\sim 59\%$ and $\sim 40\%$ of correct reconstructions. In contrast, the overall efficiency in the reconstruction of total rho events is around $\sim 80\%$ as will be shown in section table 4.2; therefore the rejection is considered valid. This filter removes around $\sim 5\%$ of the total reconstructed ρ events. This loss in statistics is perfectly reasonable given the abundance of events expected at FCC-ee and the rejection efficiency shown.

1 Photon events		2 Photon events	
MC type	Count	MC type	Count
other	15066	other	53947
μ	11667	μ	69
e	14506	e	76
π	53985	π	1285
ρ	259117	ρ	304736
a_1	87351	a_1	386483
Total	4.4×10^5	Total	7.5×10^5
Rejection Eff.	41 %	Rejection Eff.	60 %

Table 4.1: Rejection efficiency for the invariant mass filter in reconstructed ρ events. Efficiency is calculated as the fraction of non-rho background in the total events. This sample of rejected events was generated on 50×10^6 total events

The mass filter for a_1 events was also shown to be effective, with a $\sim 53\%$ good rejection rate.

4.2.3 Confusion matrix and hadronic background

One last requirement was quickly introduced when I noticed that the π channel was heavily influenced by leptonic backgrounds. In particular, below the 2 GeV

4.2 Event selection

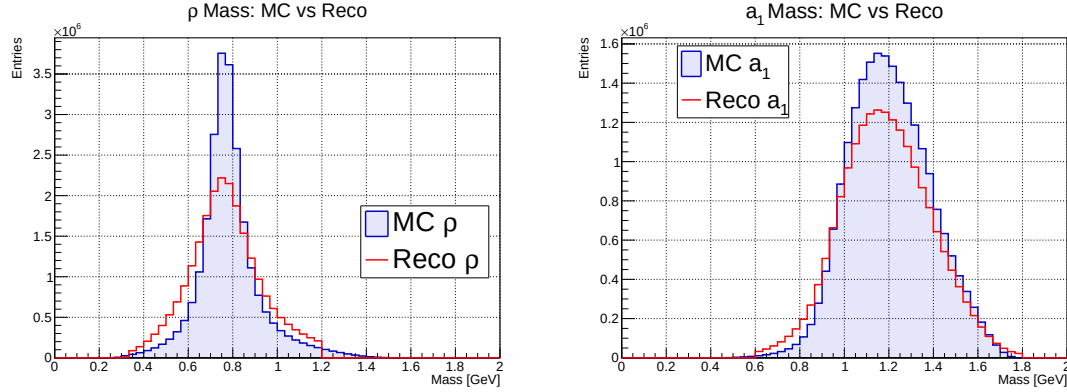


Figure 4.3: Reconstructed and Monte Carlo invariant masses for ρ and a_1 resonances. A wide mass window is identified by limiting the length of the tails.

the ParticleFlow algorithm was such that it didn't differentiate well charged traces, and therefore Monte Carlo charged leptons were attributed the pion mass in the reconstructed branch. This effect can be seen in figure 4.4. In addition to leptons migrating into pions below the 2 GeV threshold, photons were also not assigned to their own branch below this cut. This limitation adds another layer of difficulty in reconstructing hadronic events where a soft photon may be lost. It is important to notice, however, that this experimental limit is a straight consequence of the chosen characteristics in the card file representing the IDEA detector at the time of the data generation. This value was probably chosen according to the minimum momentum needed by muon and electrons to be efficiently detected in the external subsystems of the IDEA detector, but it could be lowered using more refined reconstruction techniques in the low momentum region.

This limitation forced me to filter the energy spectrum of lepton and pion events for $E > 2$ GeV, which will restrict the domain of the optimal variable considered.

When reconstructing the ρ channel no evident energy cut was needed in order to reduce background, as shown in figure 4.5.

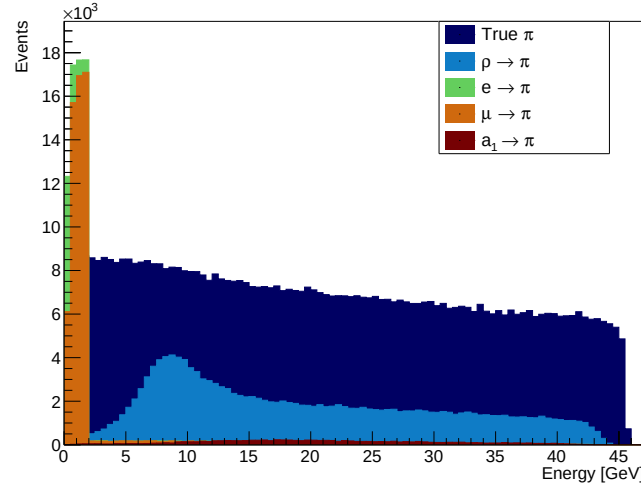


Figure 4.4: Background contributions to the events reconstructed as single π .

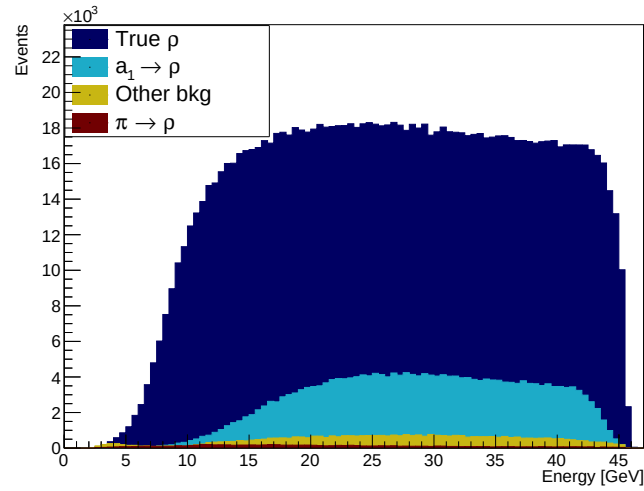


Figure 4.5: Background contributions to the events reconstructed as ρ decays.

4.2 Event selection

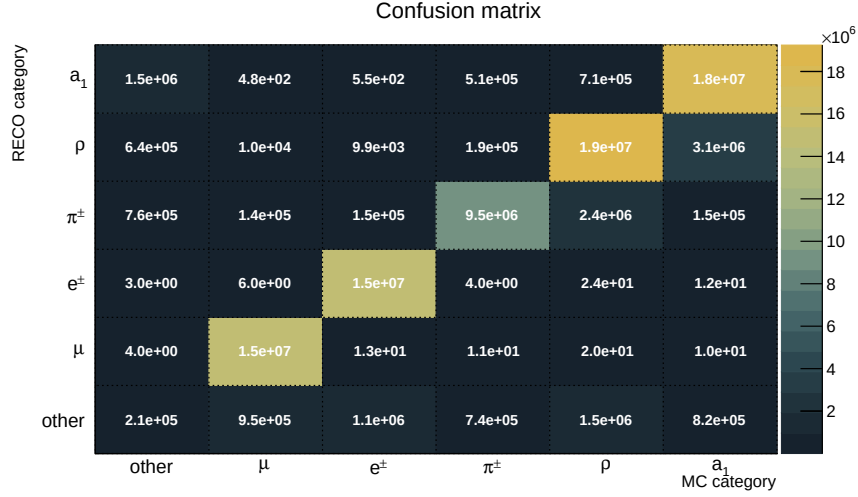


Figure 4.6: Confusion matrix constructed with Monte Carlo and Reconstructed event types. Generated from the analysis of 5×10^7 $Z \rightarrow \tau\tau$ events.

All events that do not fit the reconstructed requirements and all Monte Carlo events that differ from the chosen channels, are stored as a generic "*other*" for the purpose of this study. By combining the results of the identification process and the pre-selection cut I constructed the confusion matrices. These are double-entry tables where Monte Carlo and Reconstructed event identification are compared; this method directly shows the performance of the detector and the reconstruction process. Specifically on the horizontal axis I introduced Monte Carlo event types, while on the vertical one the Reconstructed ones (figure 4.6). Additional useful information can be obtained by normalizing each table by rows or columns. These two cases are named *efficiency* and *purity* of the reconstruction: they respectively describe the efficiency of recognizing a given Monte Carlo decay and the purity of a given Reconstructed decay channel. These results are summarized in table 4.2, while the complete matrices are found in the appendix A.1.

This data already shows how a simple reconstruction process can yield highly efficient results for reconstructing events from the main decay channel of the τ lepton.

Decay	Efficiency [%]	Purity [%]
e	93.0	100.0
μ	92.2	100.0
$\pi^\pm$	86.9	72.5
ρ	80.5	83.1
a_1	81.6	86.6
other	6.8	4.0

Table 4.2: Diagonal elements of the purity and efficiency matrices. Efficiency shows the percentage of Monte Carlo decays successfully reconstructed. Purity shows the percentage of reconstructed events correctly identified.

4.3 Templates and fits

After constructing a working identification workflow, I moved onto collecting the kinematic variables needed to perform polarization fits. Depending on the reconstructed event type I calculated the channel-dependent optimal observable.

Particular tuning had to be implemented in the ρ decay, mainly due to the calculations to reconstruct the angles in the decay. I quickly realized that some value values of $\cos \psi_\tau$ and $\cos \psi_\rho$ were getting flagged as exceeding their natural domain of $[-1, +1]$. These events were flagged as un-physical and therefore assigned the reconstructed event of "*other*". While this cut is only $\sim 25\%$ effective at background rejection, the consequent optimal variable would very likely be miss-calculated. The events flagged as such are in the order of $\sim 0.53\%$ of the total events, a negligible amount given the statistical conditions of FCC-ee.

4.3 Templates and fits

4.3.1 Histogram re-weighting

Analytical fits are not, in general, the most common choice for fitting these reconstructed distributions. This is because they can easily get distorted by kinematic approximations, but more importantly they can't include background unless it also follows a predictable distribution. This can be easily seen by fitting analytically the leptonic channels, one example in figure 4.7.

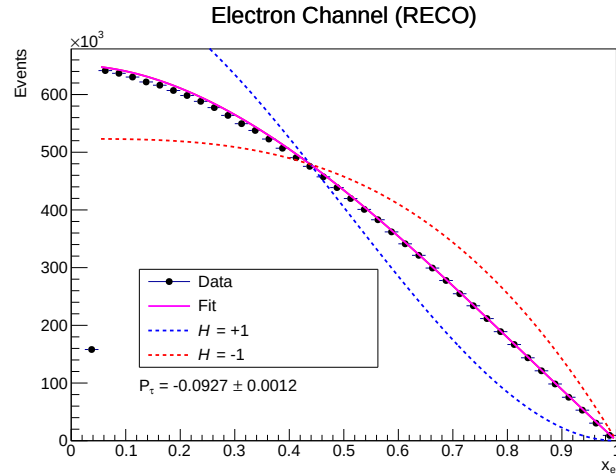


Figure 4.7: Analytical fit for the electron channel.

Being a 100% pure channel the data is well distributed on the analytical distribution; nevertheless the fit clearly doesn't report back a correct polarization value. This is because non-trivial effects, such as final state radiation by the tau, distort slightly the analytical distribution in ways that can't easily be accounted for. As seen in figure 4.4 the pion channel suffers also from an important background which has to be accounted for in other ways.

A common approach, as previously done for LEP analysis for example, is to create helicity template histograms for $h = \pm 1$ values and then fit the data onto a linear combination of those two. The process requires to take the distribution of the reconstructed events and divide them into two histograms according to their

corresponding Monte Carlo helicity; this generates two templates to be used as a basis for the linear combination. This approach was unfortunately not possible because the data that were available did not save the *helicity* value in the Monte Carlo properties. Since I did not have access to the true helicity and re-generating all of the events was not feasible, a re-weighting approach was necessary.

As previously introduced, the decay distribution given a chosen observable ξ can always be written as a linear combination of a polarization dependent and independent part:

$$W(\xi) = f(\xi) \left(1 + \mathcal{P}_{SM} \omega(\xi) \right) \quad (4.2)$$

where ω is the optimal variable and $\mathcal{P}_{SM}$ the polarization described by the Standard Model.

Given that the data were generated with PYTHIA8.3, the events are expected to be polarized according to eq 3.6 which makes use of the generator level value of $\sin^2 \theta_W = 0.2315$ which produces the values of the asymmetry parameters via eq. 3.3 and 3.8. In order to generate templates without re-running the simulation one can define a scaling factor to simulate a chosen polarization. The Standard Model distribution follows $W_{SM}(\xi) \propto (1 + \mathcal{P}_{SM} w)$, while the templates we want to generate should follow $W_{new}(\xi) \propto (1 + \mathcal{P}_{new} w)$ where $\mathcal{P}_{new} = \pm 1$ for the opposite helicity templates. The scaling factor, or histogram weight, is therefore defined as the ratio between them:

$$\Omega = \frac{W_{new}(\xi)}{W_{SM}(\xi)} = \frac{1 + \mathcal{P}_{new} w}{1 + \mathcal{P}_{SM} w} \quad \text{where } \mathcal{P}_{new} = \pm 1 \quad (4.3)$$

The non polarization-dependent part of the distribution simply vanishes since it is in common between the two.

Given the kinematic configuration of the event, a weight $\Omega > 1$ indicates that the event is more probable to have been generated by a polarization $\mathcal{P}_{new}$ rather than the baseline value. In contrast, a weight $\Omega < 1$ suggests that the event is unlikely to be from $\mathcal{P}_{new}$.

4.3 Templates and fits

This mechanism provides a tool for generating the two helicity templates for each decay channel. The procedure can be summarized in these steps:

1. Gather a sufficiently large sample for generating templates; best if the statistics are comparable to the data it will be fitted to.
2. Perform the reconstruction, classify the RECO event and calculate the RECO optimal variable.
3. For each RECO event, calculate the true Monte Carlo decay and the optimal variable by using Monte Carlo information.
4. With the Monte Carlo information, calculate the weights $\Omega^\pm$ for the two templates of $\mathcal{P}_{new} = \pm 1$.
5. Construct the two templates by filling the histograms with the reconstructed optimal variable re-weighted by the factor $\Omega^\pm$.

This process re-shapes the reconstructed optimal variable distribution into two histograms that mimic fully polarized events. While this strategy is certainly valid, it will be shown to not be completely satisfactory.

In order to polish the analysis, I ensured that the events for the templates and for the final data fit were taken from independent samples. Additionally, the samples for the templates were evenly split for the two signs $h = \pm 1$ to ensure complete independence. An example of the obtained templates can be seen in figure 4.8

Finally, the distributions of the data gathered after the event selection can be fitted onto a linear combination of the templates to obtain the polarization value. More specifically, by defining the total number of events stored $N = N_+ + N_-$ and solving for N_+ and N_- in terms of N and $\mathcal{P}_\tau$, the kinematic distribution $W(\xi)$ is obtained by fitting the linear combination

$$W(\xi) = N \left[\left(\frac{1 + \mathcal{P}_\tau}{2} \right) H_+(\xi) + \left(\frac{1 - \mathcal{P}_\tau}{2} \right) H_-(\xi) \right] \quad (4.4)$$

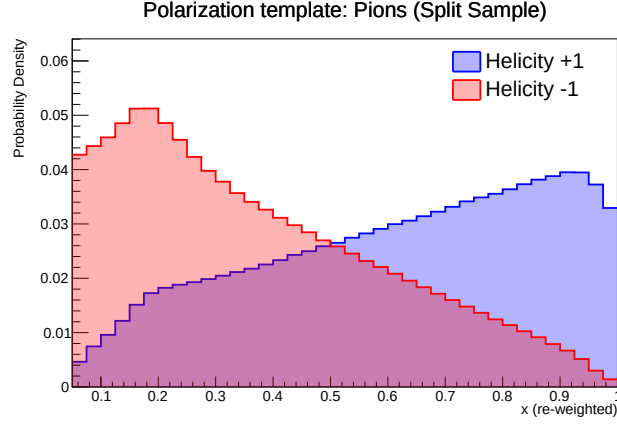


Figure 4.8: Polarization templates for the pion channel. Note that the histograms are re-scaled to unitary area in order to represent a probability density.

The total number of events is fixed by the given histogram integral, and therefore the ROOT minimization process has to fit for only one free parameter: $\mathcal{P}_\tau$. The fitting method used is a *binned Log-Likelihood*, already implemented in the ROOT libraries via the option L. This is the standard procedure given the Poissonian nature of the event distribution per bin, and also to handle any low count bins.

Chapter 5

Polarization estimates

In this last chapter I will provide the results obtained and the methodologies used to estimate the errors involved.

The *FCC Feasibility Study* report [8] estimates 6×10^{12} visible Z decays at FCC-ee with an integrated luminosity of $\mathcal{L}_{\text{int}} = 140 \text{ ab}^{-1}$ (see tab. 1.1). Given the branching ratios [11], the expected number of $Z \rightarrow \tau\tau$ events is calculated via

$$N_{Z \rightarrow \tau\tau} = N_Z \times BR_{Z \rightarrow \tau\tau} = 2.02 \times 10^{11} \quad (5.1)$$

The channel-specific branching ratios shown in table 5.1 yield a number of expected events per channel

$$N_x = 2 \times N_{\tau\tau} \times BR_x \quad (5.2)$$

Given the number of events, the expected precision at FCC-ee can then be extrapolated from the sensitivity calculated from the fits performed.

As previously mentioned, the data are expected to be generated using the PYTHIA8.3 default value $\sin^2 \theta_W^{eff} = 0.2315$. Given the asymmetry parameters from eq. 3.3, the expected average polarization is $\langle \mathcal{P}_\tau \rangle = -\mathcal{A}_\tau = -0.1472$.

The following sections provide the results obtained from this study in the various decay channels. The main analysis has been run on independent samples of 5×10^7

5.1 Leptonic channels

Decay mode	BR [%]	N _{FCC-ee}
$\tau \rightarrow e\bar{\nu}\nu_\tau$	17.82 ± 0.04	7.2×10^{10}
$\tau \rightarrow \mu\bar{\nu}_\mu\nu_\tau$	17.39 ± 0.04	7.0×10^{10}
$\tau \rightarrow \pi\nu_\tau$	10.82 ± 0.05	4.4×10^{10}
$\tau \rightarrow \rho\nu_\tau$	25.49 ± 0.09	10.3×10^{10}

Table 5.1: Expected per channel number of events at FCC-ee.

total events for both the templates and data. I must stress that the dataset used contains only $Z \rightarrow \tau\tau$ events, and therefore only internal background is included. I recall here for your convenience the definition of the histogram weights used in re-weighting:

$$\Omega = \frac{W_{new}(\xi)}{W_{SM}(\xi)} = \frac{1 + \mathcal{P}_{new}\omega}{1 + \mathcal{P}_\tau\omega} \quad (5.3)$$

where $\mathcal{P}_{new} = \pm 1$ and ω is the optimal variable.

5.1 Leptonic channels

Leptonic channels are fitted via the re-weighting technique, making use of the kinematic variable $x_{lep} = E_{lep}/E_\tau$. Section 4.2.3 showed how for $E < 2 \text{ GeV}$ the detector limitations introduced a prominent migration between leptons and hadrons. Aware of this issue, the leptonic kinematic variable was limited by the corresponding value of $x > 2/45.5 \sim 0.05$. Consequently, also the templates were normalized to unity and generated only in the corresponding domain of $x \in [0.05, 1]$.

The x variable was extracted by simply collecting the lepton energy in the corresponding reconstructed events, and dividing it by the assumed tau energy $E_\tau = \sqrt{s}/2 \simeq m_Z/2 \simeq 45.594 \text{ GeV}$. It must be noted that radiative effects from the decayed tau can lower its energy. These events are partially accounted for since the templates are generated by using a re-weighted distribution of reconstructed data, which has the same bias as the data fitted.

Leptonic channels				
Type	Template events	Data events	$\langle \mathcal{P}_\tau \rangle$	χ^2/ndf
e	7.31×10^6 each	1.45×10^7	-0.1585 ± 0.0012	117/37
μ	7.23×10^6 each	1.43×10^7	-0.1591 ± 0.0012	59/37

Table 5.2: Obtained fit results for reconstructed leptonic decays. Analysis of an overall 5×10^7 event sample.

The weights for histogram re-weighting (see eq. 4.3) make use of the optimal variable $\omega = b(x)/a(x)$, where $a(x)$ and $b(x)$ are the functions defined for the decay distribution 3.17. Recall that the optimal variable is shown to be fully equivalent to x for leptonic decays, therefore the variable plotted and fitted is x , given its simplicity. Full results are shown in table 5.2 and figure 5.1.

5.2 Pion channel

The pion decay is fully described, in analogy to the leptonic ones, by the sole kinematic variable $x_\pi = E_\pi/E_\tau$. Given the definition, it suffers from the same issues described in the leptonic section above. In contrast, the optimal variable used in the re-weighting is different.

Recall that hadronic decays can be described by

$$\frac{dN}{d \cos \theta^*} = \frac{1}{2}(1 + \mathcal{P}_\tau \alpha \cos \theta^*) \quad (5.4)$$

where θ^* is the decay angle between the pion and the tau in the tau rest frame. In the case of pions $\alpha = 1$, which means full sensitivity is recovered by re-weighting with the variable $\omega = \cos \theta^*$. Full results are shown in table 5.3 and figure 5.2

It is important to notice that the pion channel is mainly distorted by the migration of ρ events. These events were re-weighted without recovering full sensitivity, as will

5.2 Pion channel

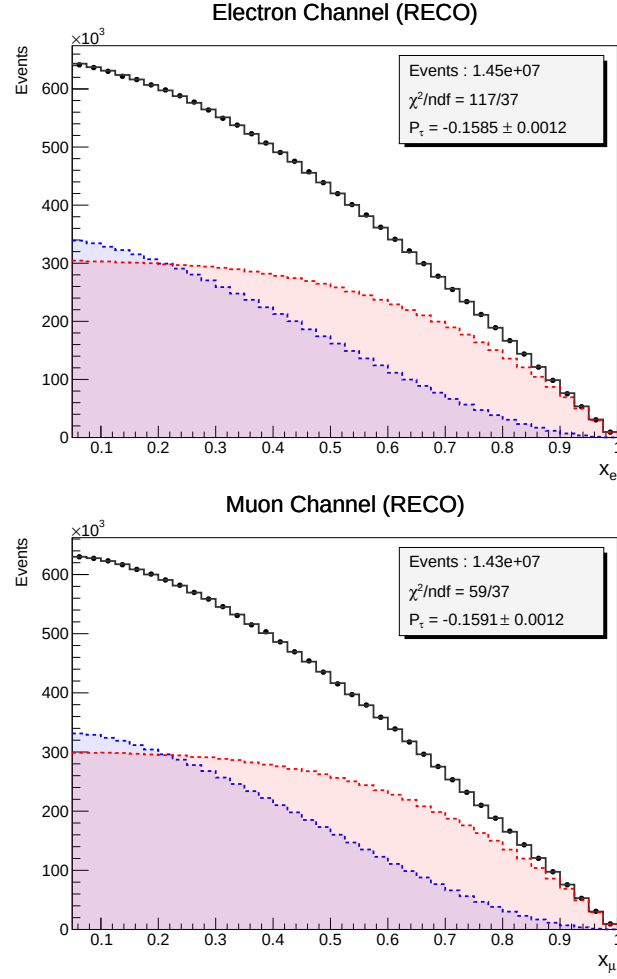


Figure 5.1: Final fits of leptonic channels with templates generated by re-weighting. Red and blue shaded areas represent respectively $h = -1$ and $h = +1$ templates.

Pion channel				
Type	Template events	Data events	$\langle \mathcal{P}_\tau \rangle$	χ^2/ndf
π	7.77×10^6 each	1.31×10^7	-0.1538 ± 0.0006	229/37

Table 5.3: Obtained fit results for reconstructed pion decays. Analysis of an overall 5×10^7 event sample.

be motivated in the section below. This fact yields a partial loss in the measured sensitivity.

5.3 Rho channel

Rho channel events were assigned their optimal variable ω_ρ by the definition

$$\omega = \frac{W^+(\xi) - W^-(\xi)}{W^+(\xi) + W^-(\xi)} \quad (5.5)$$

where $W^\pm$ are the positive and negative helicity decay distributions defined as eq. 3.20, and the involved kinematic variables are the angles $\psi_\rho, \psi_\tau, \eta$ defined in eq. 3.27, 3.21, 3.26.

Partial sensitivity is obtained by considering the generic expression of hadronic decays, recalled in the previous section for pion decays. For rho decays the coefficient yields $\alpha = \frac{m_\tau^2 - 2m^2}{m_\tau^2 + 2m^2}$, which means that by constructing histogram weights as

$$\Omega^\pm = \frac{1 \pm \alpha \cos \theta_\rho}{1 + \mathcal{P}_\tau \alpha \cos \theta_\rho} \quad (5.6)$$

one can expect to recover on average a factor of $\alpha \sim 0.46$ of the fully polarized weight for true rho events.

Significant sensitivity can be recovered by re-weighting with the optimal variable ω . This method is expected to boost the sensitivity from 0.26 to ~ 0.49 (see table 3.2), since it uses more kinematic information. While this would be the optimal

5.3 Rho channel

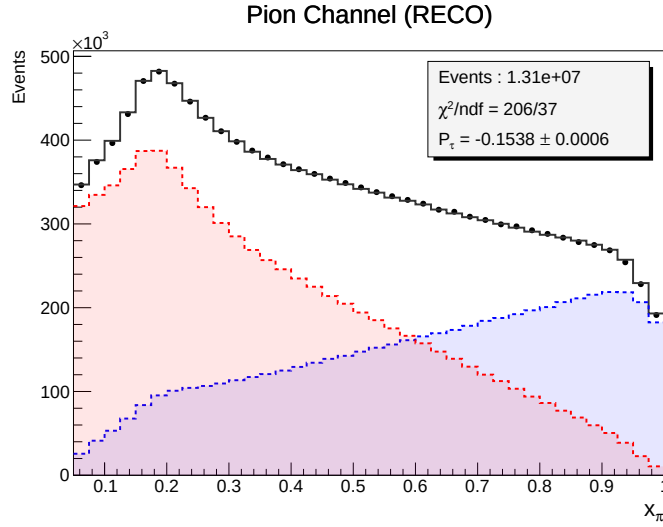


Figure 5.2: Final fit for pion channel with templates generated by re-weighting. Red and blue shaded areas represent respectively $h = -1$ and $h = +1$ templates.

choice, when this technique was implemented I encountered multiple issues with the fit results for both rho and pion decays. This can be attributed to the fact that the pion decay suffers from a prominent, and difficult to remove, rho background (see figure 4.4). The rho decay on the other hand sees a migration of a_1 events (see figure 4.5). These migrations distort the decay distributions in important ways. Because the optimal variable is constructed precisely to maximize the differentiation of negative and positive helicity events, it is highly sensitive to background events.

To ensure a more stable and robust approach, the analysis implemented the $\cos \theta^*$ weight. While it has a lower analysing power, it allowed me to proceed with consistent fits. The results obtained are shown in table 5.4 and figure 5.3.

Rho channel				
Type	Template events	Data events	$\langle \mathcal{P}_\tau \rangle$	χ^2/ndf
ρ	1.15×10^7 each	2.29×10^7	-0.1430 ± 0.0016	64/39

Table 5.4: Obtained fit results for reconstructed rho decays. Analysis of an overall 5×10^7 event sample.

5.4 Fit validation and FCC-ee projection

In order to estimate the statistical fluctuation of the fitting technique, another analysis was performed: I proceeded by generating templates with 20×10^6 events, while the remaining 80×10^6 events were then analysed in 80 independent samples of 1×10^6 events each. The idea is to have stable templates and repeat the fitting technique over 80 data samples. The results from those 80 fits are stored and the standard deviation (σ_{1M}) of the extracted value for each channel provides an estimate of the fitting fluctuations. The results obtained are shown in figure 5.4.

Results from the global fit over 50M events are then compared to the repeated analysis over $80 \times 1M$ events samples. In a consistent analysis the uncertainties are expected to respect the proportionality $\sigma \sim 1/\sqrt{N}$, meaning that

$$\sigma_{1M} \sqrt{1 \times 10^6} \sim \sigma_{50M} \sqrt{50 \times 10^6} \quad (5.7)$$

Therefore the expected scaling factor is

$$\sigma_{1M} \sim \sqrt{50} \times \sigma_{50M} \sim 7.1 \times \sigma_{50M}$$

The observed result is that the fitting technique scales appropriately with the sample size of data; a complete comparison is reported in table 5.5.

After confirming the validity of the error estimation, the final results can be shown. The estimated value for $\sin^2 \theta_W$ is given by combining $\langle \mathcal{P}_\tau \rangle = -\mathcal{A}_\tau$ and the equations

$$\mathcal{A}_\tau = 2 \frac{g_V/g_A}{1 + (g_V/g_A)^2} \quad \text{and} \quad \frac{g_V}{g_A} = 1 - 4 \sin^2 \theta_W \quad (5.8)$$

5.4 Fit validation and FCC-ee projection

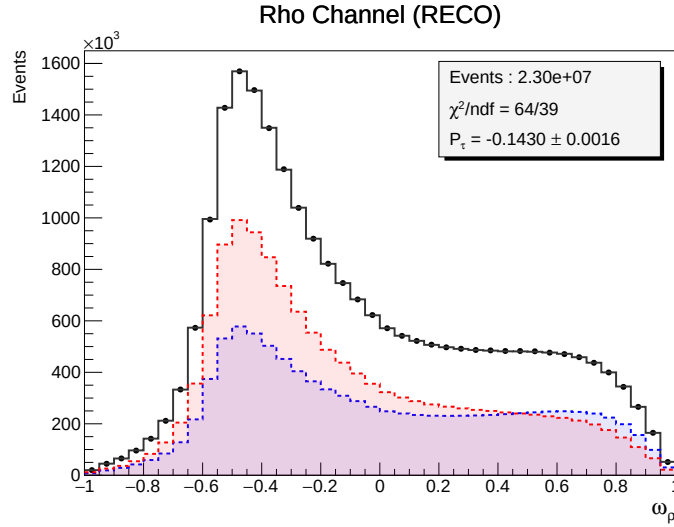


Figure 5.3: Final fit for rho channel with templates generated by re-weighting. Red and blue shaded areas represent respectively $h = -1$ and $h = +1$ templates.

Decay	Std Dev (80 samples) (1×10^6 events)	Global fit error (5×10^7 events)	Scaling factor
e	0.0080	0.0012	6.7
μ	0.0078	0.0012	6.5
π	0.0037	0.0006	6.2
ρ	0.0118	0.0016	7.4

Table 5.5: Validation of the error estimated in the fitting technique.

Full results are shown in table 5.6

Only the ρ channel results compatible with the Standard Model value, while the other decays suffer from major systematic errors; these are partially addressed in the next section 5.5. Given the non reliable results obtained, the combined values are to be taken as illustrative.

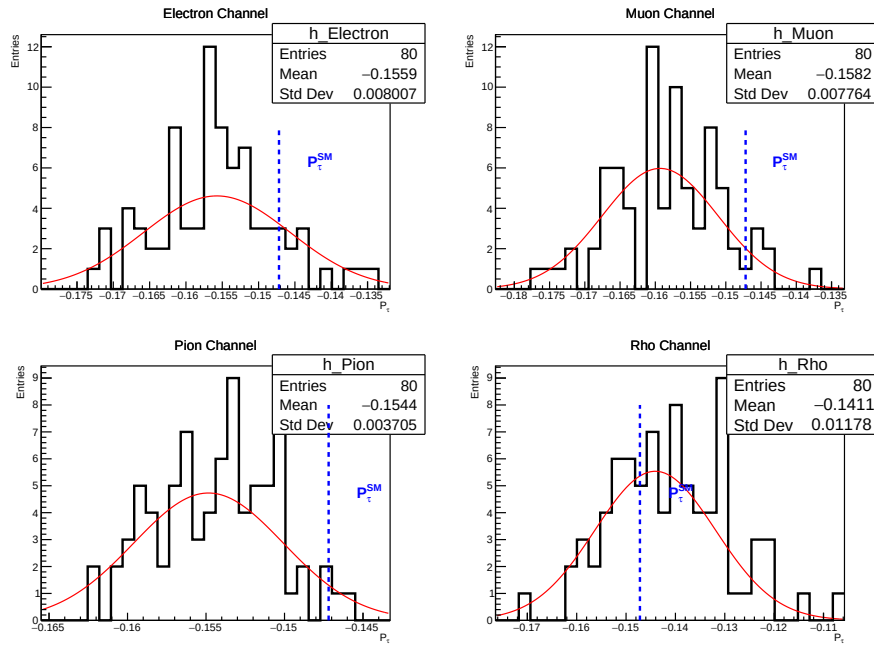


Figure 5.4: Fitted values of $\langle \mathcal{P}_\tau \rangle$ in the different channels, across the 80 independent analysis of 1×10^6 event samples. The blue dotted line represent the generator level value of $\langle \mathcal{P}_\tau \rangle$ according to PYTHIA8.3 .

5.4 Fit validation and FCC-ee projection

Global fit estimated values			
Decay	$\langle \mathcal{P}_\tau \rangle$	g_V/g_A	$\sin^2 \theta_W$
e	$-0.1586(12)$	$0.0798(6)$	$0.2300(2)$
μ	$-0.1585(12)$	$0.0797(6)$	$0.2301(2)$
π	$-0.1538(6)$	$0.0771(3)$	$0.23073(8)$
ρ	$-0.1430(16)$	$0.0719(8)$	$0.2320(21)$
Combined	$-0.1540(5)$	$0.0774(2)$	$0.23063(6)$
PYTHIA8:	-0.1472	0.0740	0.2315

Table 5.6: Obtained results for electro-weak coupling constants. Analysis of an overall 5×10^7 event sample.

Finally, the per-channel sensitivity can be obtained simply via

$$S_i = \frac{1}{\sigma_i \sqrt{N_i}} \quad (5.9)$$

where N_i and σ_i are the reconstructed number of events per channel and the statistical uncertainty obtained. The expected number of events, table 5.1, and the uncertainties from table 5.6 are then used to extract the expected precision available at FCC-ee for the measurement of $\mathcal{P}_\tau$ and $\sin^2 \theta_W$.

Expected FCC-ee precision			
Decay	$S_{\mathcal{P}_\tau}$	$\sigma(\mathcal{P}_\tau)$	$\sigma(\sin^2 \theta_W)$
e	0.22	1.7×10^{-5}	2.2×10^{-6}
μ	0.22	1.7×10^{-5}	2.2×10^{-6}
π	0.46	1.0×10^{-5}	1.3×10^{-6}
ρ	0.13	2.5×10^{-5}	3.3×10^{-6}
Combined	0.26	7.1×10^{-6}	9.0×10^{-7}

Table 5.7: Expected precision of the measurements at FCC-ee.

The sensitivities can be compared to an earlier Monte Carlo study [17], which did

not include reconstruction effects and the consequent cross-channel contaminations. Leptonic decays, given the 100% purity of reconstruction, maintain full sensitivity at 0.22. Pion decays in this study suffer from a prominent ρ background which lowers the sensitivity ($0.57 \rightarrow 0.46$). Rho decays instead suffer from a_1 backgrounds, which were also re-weighted in a non-optimal way. Therefore the rho channel suffers the most the reconstruction effect and its sensitivity drops from 0.49 to 0.13.

5.4.1 Leptonic universality

Lastly, a test for leptonic universality was performed by fitting the angular distribution described by equation 3.6, which defines the polarization $\mathcal{P}_\tau$ as a function of the angle $\cos\theta$ of the tau decay with respect to the beam axis. This measurement allows for the independent estimation of the two asymmetry parameters $\mathcal{A}_\tau$ and $\mathcal{A}_e$, a powerful tool for testing leptonic universality. The angle for each reconstructed event was taken as the thrust direction in the respective hemispheres. The fit was performed for each channel over the reconstructed domain of $|\cos\theta| < 0.95$. Full results are shown in table 5.8.

Leptonic universality		
Decay	$\mathcal{A}_\tau$	$\mathcal{A}_e$
e	0.1626(17)	0.1396(19)
μ	0.1636(17)	0.1404(19)
π	0.1530(10)	0.1130(11)
ρ	0.1433(20)	0.1375(24)
Combined	0.1544(8)	0.1230(9)
PYTHIA8:	0.1472	0.1472

Table 5.8: Test for leptonic universality by fitting $\mathcal{P}_\tau(\cos\theta)$.

Given the difficulties encountered, as described in prior sections, the results obtained are not satisfactory and are not compatible with the generator value of

5.5 Systematic errors

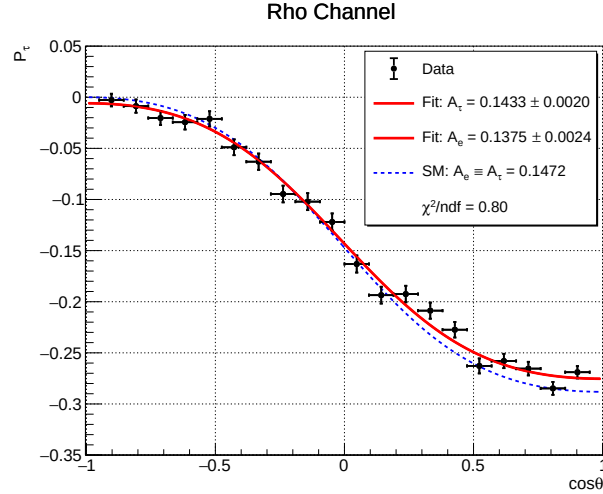


Figure 5.5: $\mathcal{P}_\tau$ as a function of $\cos\theta$ in the ρ channel. The plot allows to independently fit for the two asymmetry parameters.

PYTHIA8.3. In the appendix A.5 various plots for different bins of $\cos\theta$ are reported; these show that while the fitted values are not compatible, the technique is viable and the templates are correctly re-shape throughout the angular spectrum.

5.5 Systematic errors

In this paragraph I wish to partially address some of the incompatibilities observed in the results. A more refined study should be implemented when fully polarized templates will be available for analysis, in order to exclude with more confidence the impact of the fitting techniques.

In an attempt to investigate the polarization obtained in leptonic decays, I analysed a possible systematic underestimation of the kinematic variable $x = E_{lep}/E_\tau$. Possible effects come from final state radiation by the tau or Bremsstrahlung from electrons.

In practice I analysed the factor

$$\frac{\Delta x}{x} = \frac{x_{reco} - x_{MC}}{x_{MC}} \quad (5.10)$$

to check for any systematic energy shift. No meaningful systematic shift was observed in a sample of 10×10^6 events. The plotted distributions are reported in the appendix A.7.

In pion decays, the rho background has an explicit energy peak at low reconstructed energies with respect to E_{beam} (fig. 4.4). It's important to notice that in this area the negative helicity template dominates. Since the kinematic variable for the pion channel is exactly proportional to the reconstructed charged energy, the background introduces an important systematic effect. Although the re-weighting procedure should naturally handle background, given the strong presence it could still impact the fitting performance.

In contrast, the a_1 background for reconstructed ρ events, is distributed uniformly along the optimal variable ω_ρ (see appendix A.3). Together with the lower sensitivity to polarization (0.12), the background dilutes the signal, resulting in a worse precision but a less biased measure.

Final remarks

The latest FCC review [2] identifies a precision of 5×10^{-6} as the target for the measurement of $\sin^2 \theta_W$; this analysis produces an expected precision of 9×10^{-7} . An earlier study [17] performed only over Monte Carlo data, also demonstrated a possible precision of 5×10^{-7} . Considering the loss of sensitivity in hadronic decays and the inclusion of cross-channel contaminations, the official FCC-ee target is well in reach. It should also be noted that these results are derived from a re-weighting technique, which has a lower statistical efficiency compared to generating templates from fully polarized datasets.

Loss in sensitivity, especially in hadronic decays, is driven by the issues encountered when implementing re-weighting with the fully optimal variables. Instabilities in the fitting technique require further study, particularly a better background removal. In a future reiteration of the Monte Carlo production, fully polarized samples will ensure full reliability of the templates providing the kinematic distributions, without having to perform a re-weighting a posteriori.

Results from the angular distribution $\mathcal{P}_\tau(\cos \theta)$ show incompatibilities in the polarization estimates, especially near the outer bounds of $\cos \theta \rightarrow \pm 1$. These issues require further investigation once the systematic errors in non-rho channels have been addressed.

Appendix A

Additional material

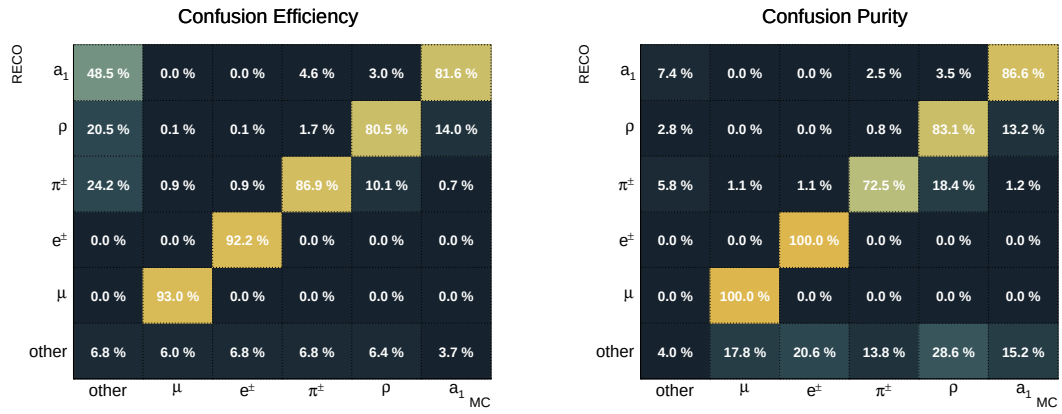


Figure A.1: Complete confusion efficiency and purity matrices.

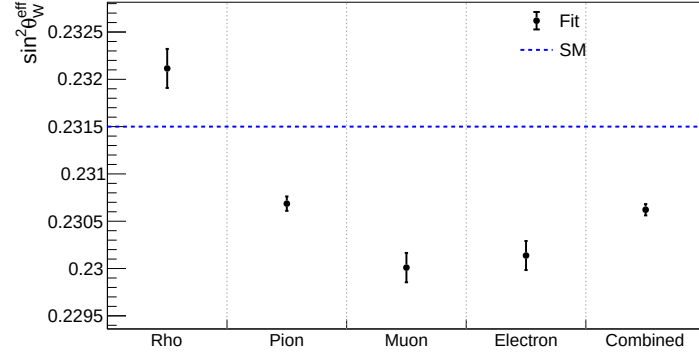


Figure A.2: Summary of the obtained results for $\sin^2 \theta_W^{eff}$

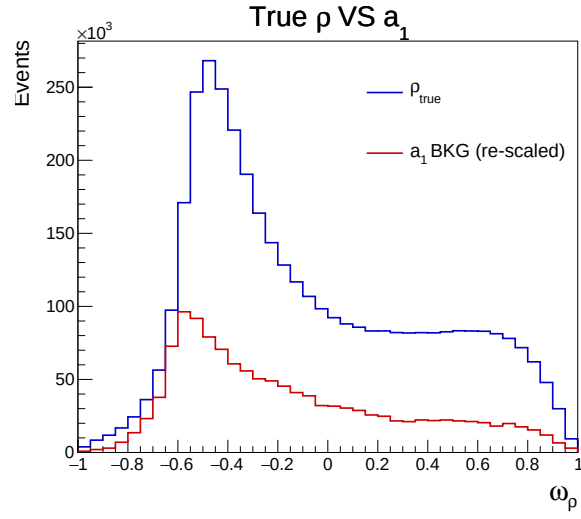


Figure A.3: Contributions of true ρ and a_1 background (re-scaled).

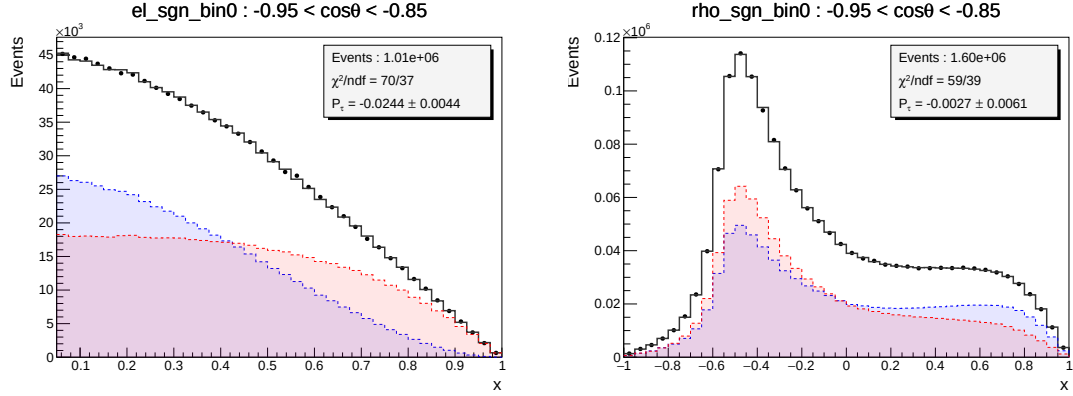


Figure A.4: $\mathcal{P}_\tau$ in the first bin $-0.95 < \cos \theta < -0.85$ for ρ and e decays. Notice the similar areas covered by the two helicity templates, signaling an average null polarization.

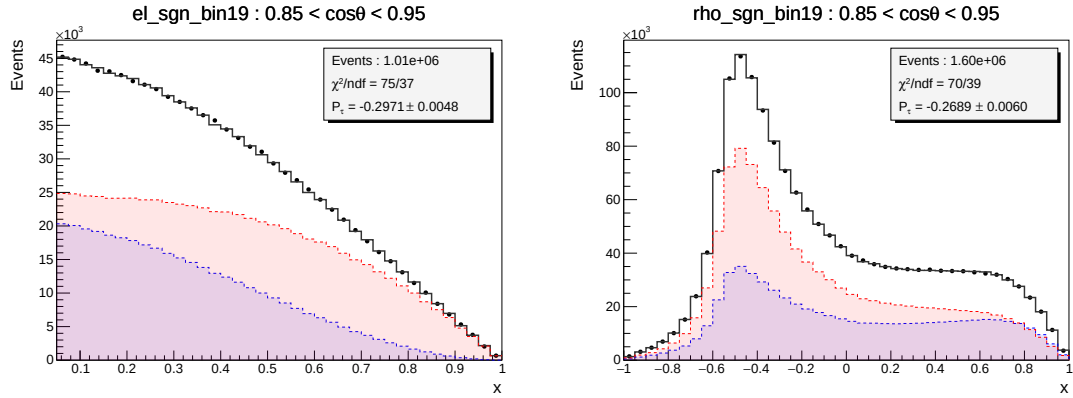


Figure A.5: $\mathcal{P}_\tau$ in the last bin $0.85 < \cos \theta < 0.95$ for ρ and e decays. Notice the change in the template areas.

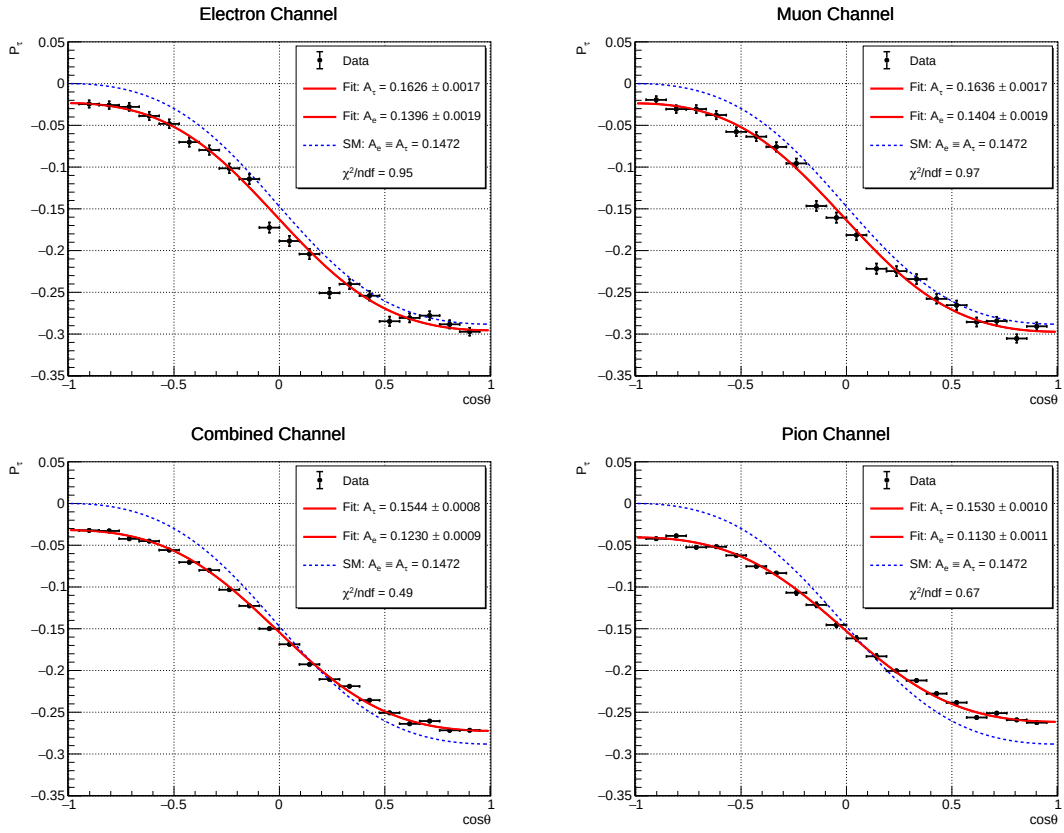


Figure A.6: Example of leptonic universality tests for the remaining decays, and a combined test.

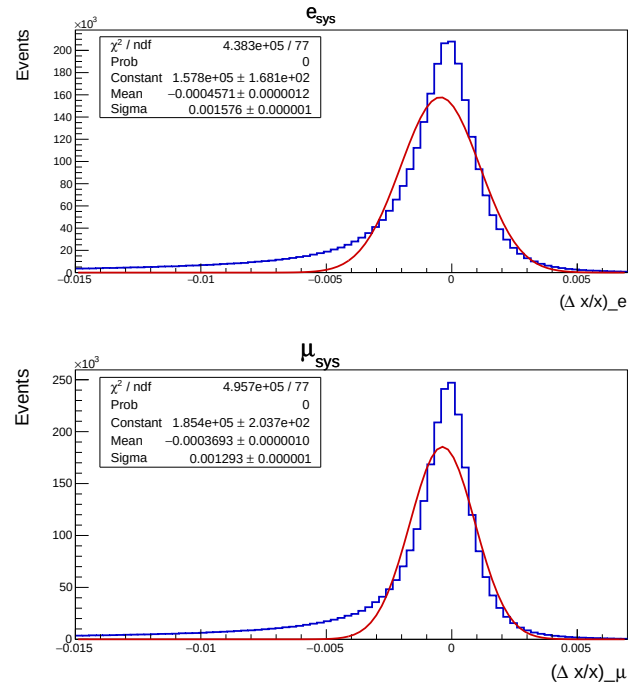


Figure A.7: Distribution of the $\Delta x/x$ calculated as an estimate of systematic error in leptonic decays.

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